Drawing Credible Directed Acyclic Graphs for Causal Inference

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Abstract

Causal directed acyclic graphs (DAGs) are intelligible representations of real-world data-generating processes that facilitate causal inference by providing (automatized) guidance for assessing whether a causal effect is identified with the observed data and for selecting covariates that remove most, if not all confounding bias. However, less attention has been paid to the process of *constructing* causal DAGs. Methodological work often relies on toy examples that have limited practical utility for applied researchers working in complex contexts. This article introduces and demonstrates a stepwise, iterative procedure for drawing credible causal DAGs, which is designed to guide researchers in identifying important sources of confounding while also incorporating research design features of quasi-experiments or random experiments, as well as threats to validity (e.g., measurement error, treatment non-compliance). Although constructing a complete DAG that fully captures the data-generating process is difficult and rarely achievable in practice, we argue that developing a *credible* DAG—one that includes all plausible sources of confounding—is adequate for applied research. The proposed iterative drawing procedure is directly aligned with the goal of constructing credible causal DAGs.

Keywords: directed acyclic graphs, causal inference, data-generating process, confounding, validity threats

Causal inference is necessary for evaluating interventions and understanding the mechanisms through which they produce their effects. However, causal inference is also challenging because "data are profoundly dumb about causal relationships" (Pearl & Mackenzie, 2018, p. 14). In other words, data on their own do not contain any information about the causal processes that generated them, which makes it impossible to isolate the effects of interventions from data alone. At best, data provide correlations—probabilistic associations between variables—which do not necessarily describe what would happen if one were to actively intervene on a variable because correlations do not imply causation. What additional knowledge, then, is required to derive a causal estimate from the correlations that data provide? Quantitative methodologists and philosophers agree that we need qualitative background knowledge about the *data-generating process* (DGP; Pearl, 2009; Spirtes et al., 2000; Woodward, 2003): knowledge about the processes that produce the observed data, which in the social sciences include the study participants' (i) characteristics like attitudes, behaviors, or traits, (ii) how they interact with the research design and measurement instruments, and (iii) other idiosyncratic causes. Without reliable knowledge of the causal processes that generated the data, causal inference is impossible. As Cartwright (1994, Chapter 2, p. 39) put it, "no causes in, no causes out."

A growing number of researchers across disciplines are acknowledging the importance of causal background knowledge in making sense of data, which represents a departure from the positivist "data-driven" paradigms that have historically dominated statistics since at least Pearson (1892), and which continue to be reflected in some areas of modern machine learning and artificial intelligence (Andrews, 2024; Boon, 2020; Nickles, 2020). Cunningham (2021, p. 117) criticizes this positivist approach and argues that

> "[a]n atheoretical approach to empiricism will simply fail. Not even 'big data' will solve it. Causal inference is not solved with more data […]. Causal inference requires knowledge about the behavioral processes […]. Without them, one cannot hope to devise a credible identification strategy."

The background knowledge about the DGP is typically expressed in the form of *causal assumptions*, which can be formalized and linked to statistics using different causal frameworks. Three of the most commonly used frameworks in the social sciences are (i) Campbell's typology of validity threats (Shadish et al., 2002), (ii) the Rubin Causal Model and its potential outcomes notation (Imbens & Rubin, 2015), and (iii) causal graphs, also known as causal directed acyclic graphs (DAGs) (Pearl, 2009; for a comparison of these frameworks see Imbens, 2020; West & Thoemmes, 2010; Steiner et al., 2023). In this article, we focus on causal graphs as a tool to formalize the causal assumptions because they offer clear advantages that have yet to be fully leveraged in the social sciences.

The main goal of this article is to provide guidance to applied researchers on how to systematically construct causal graphs that can be used for formal graphical analyses. First, we briefly discuss the importance of DAGs for causal inference and why they are not yet more widely adopted among applied researchers. We then demonstrate the process of drawing graphs using a running example, and afterwards we show how the causal DAG changes under alternative research designs and in the presence of different validity threats, highlighting their flexibility and implications for causal identification. We also briefly discuss some issues that specifically pertain to the construction of causal mediation graphs. Finally, we end with considerations about when the DGP represented in a causal graph can be considered credible to ensure valid causal inference. A brief introduction to graph terminology and basic concepts is contained in the Appendix (for more thorough introductions, see Elwert, 2013; Feng, 2024; Hernán & Robins, 2023; Huntington-Klein, 2022; Morgan & Winship, 2015; Pearl, 2010a; Rohrer, 2018; Steiner et al., 2023)

Why use causal graphs?

The primary purpose of a causal DAG is to facilitate causal inference about the impact of a *treatment* (or intervention) on an *outcome*. The process of causal inference typically involves four steps (see also Grosz et al., 2020; Heckman & Pinto, 2022): (i) formalizing qualitative background knowledge about the DGP, (ii) defining the causal estimand like the average causal effect or direct/indirect effects for mediation, (iii) conducting identification analyses, and (iv) carrying out estimation and inference. Our

focus in this manuscript is on step (i)—the first, arguably most crucial, and yet often most overlooked step in causal inference.[1]

Causal DAGs are graphical representations of the researchers' assumptions about the DGP: the causal relationships describing how the variables (observed and unobserved) affect the treatment selection and observed outcome. Once a DAG has been established, it allows researchers to formally assess whether a causal estimand of interest is *identified*, and therefore estimable from the observed data. Specifically, graphs impose a causal structure on the data which enables researchers to reason about the effects of interventions on an outcomes.[2]

Causal graphs offer a number of advantages that facilitate credible causal inference. First, given a causal graph, a set of graphical rules allows researchers to automatically determine (i) whether the causal effect is identified for the observed data and (ii) which covariates sets succeed (or fail) in removing the entire confounding bias. That is, covariate selection for causal inference is automatized without relying on empirical model selection statistics like adjusted R-squares, information criteria, or likelihood ratio tests (goodness of fit is neither necessary nor sufficient for causal inference; Spanos, 2022). Without using a causal graph, it is difficult for researchers to defend their selection of covariates and to assess whether they successfully remove all the confounding bias. Inadequate variable selection may not only fail to remove a major part of the bias but can even induce collider bias and amplify any remaining bias (Pearl, 2010b; Steiner & Kim, 2016; Wooldridge, 2009; Wright, 1921; Wysocki et al., 2022).

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[1] To make the manuscript more useful for applied researchers, we also briefly discuss aspects of (ii) defining the causal estimand and (iii) conducting an identification analysis. Also note, that we focus only on covariate adjustment methods for causal identification. A discussion of estimation methods like regression, matching, or propensity score estimators is beyond the scope of this contribution (see Imbens & Rubin, 2015; Schafer & Kang, 2008; Steiner & Cook, 2013).

[2] Psychologists are often more familiar with structural equation models (SEMs)—which, like DAGs, trace their history back to the work of Sewall Wright (1921)—as a method for decomposing correlations into structural and non-structural components based on the theory expressed in a path diagram. But unlike SEMs, DAGs are fully nonparametric and thus do not depend on the stringent and often unrealistic assumptions of linear and constant effects across all individuals. Due to their nonparametric nature, DAGs automatically encode all possible interaction and moderation effects—they do not need to be separately drawn. DAGs also go further than SEMs by drawing *theory* about the *data-generating process*, which also includes unobserved variables, whereas SEMs are limited to drawing *data*, that is, the (linear) relations between observed variables and also between latent variables if they are represented by a set of observed indicators.

Second, while these automatized procedures were devised for identification strategies that remove confounding bias via covariate adjustment (e.g., regression adjustments, propensity score matching, inverse-probability-of-treatment weighting), causal graphs also guide researchers in choosing between alternative identification strategies. For example, if the data contain a pretest measure of the outcome, causal graphs help determine whether the causal assumptions of difference-in-differences (gain score) or instrumental variable estimation are more likely satisfied than the causal assumptions for covariate adjustment methods (Kim & Steiner, 2021a, 2021b; Sheehan et al., 2024).

Third, even if the graphical analysis suggests that a causal effect cannot be identified for the given data—which is a possibility that must always be taken seriously—causal graphs can still help in assessing the direction and relative magnitude of the remaining bias and guide sensitivity analyses (VanderWeele & Robins, 2010; though research about how to use DAGs for sensitivity analysis is still missing). Finally, because causal graphs make the most important causal assumptions about the DGP explicit, they require researchers to clearly defend them. Visualizing the DGP also allows other experts and researchers to understand and criticize the assumptions encoded in the graph and potentially suggest an alternative graph that is more plausible.

Overall, causal graphs help applied researchers shift their focus from a data- and analysis-oriented thinking to a causal- and theory-driven reasoning, from often abstract statistical assumptions to intuitively understandable causal assumptions. Graphs facilitate causal inference because they provide the conceptual tools that establish a novel way of reasoning about causal inference (de Regt, 2017; Krämer, 2016; Morgan, 2012; Pearl, 2009; Spirtes et al., 2000). In discussing the epistemological value of diagrams in general, Krämer (2016, p. 83; own translation) states that “diagrams are graphic thinking tools” that “enable us to do something practically or theoretically that is difficult or impossible to do without a diagram.” Causal graphs of the DGP embody this idea by allowing us to (i) formally define causal effects, (ii) derive testable implications to assess the graph’s empirical adequacy, (iii) determine whether a causal effect is identifiable and which covariate sets suffice to remove confounding bias, (iv) help to evaluate the direction and potential magnitude of any remaining bias, and (v) clearly communicate

and defend the underlying causal assumptions. In this sense, the ability to actively and formally *use* the qualitative background knowledge encoded in causal graphs provides a more fruitful framework for causal inference than the potential outcomes approach, which lacks comparable formalized tools for integrating qualitative, context-specific background knowledge into quantitative analysis.[3] However, the potential of causal graphs can be fully realized only if researchers possess the skills to read, understand, and construct credible DAGs. This article focuses on developing the skills needed to draw causal DAGs and briefly illustrates their application.

Why are causal graphs not yet used in the behavioral and social sciences?

Despite their strengths, causal graphs have so far been rarely used in applied research in the social sciences.[4] There are two main barriers that discourage applied researchers from using causal graphs. First, causal graphs require a lot of causal information about the DGP. To draw valid inferences from a causal graph that apply to real-world settings, the graph must correctly specify *all common causes* between each pair of variables (and not only between the treatment and outcome variable).[5] The omission even of a single common cause results in a DAG that is no longer *causal*, so any causal claims derived from it may be invalid. That is, the graphical analysis may suggest that the causal effect is identified from the observed data when it actually is not. Researchers may therefore become skeptical about whether it would ever be possible to draw a realistic causal graph and instead rely on alternative frameworks that do not

[3] This argument aligns with de Regt's (2017) call for intelligible theories and general principles that allow for pragmatic understanding. According to de Regt (2017, p. 102), a theory has a high degree of intelligibility if scientists "can recognize qualitatively characteristic consequences of [the theory] without performing exact calculations." He also argues that the qualitative assessment of a theory's consequences and its use are further fostered if it provides causal explanations and can be visualized. We argue that causal DAGs constitute such intelligible representations of local theories about the data-generating process (DGP); that is, they offer a visual and causal explanation of how the data were causally generated and, importantly, can be used in a formal and systematic way for the tasks (i)-(v) listed above.

[4] In the health sciences, DAGs are already more frequently used in publications (see Tennant et al., 2021, for a review). There are a few published exceptions in other fields like sociology and economics such as Wodtke et al., (2011), Grosz et al., (2020), Huntington-Klein (2022), or causal graphs for disparity research (Howe et al., 2022). But even in these examples, the graphs often represent stylized scenarios rather than the underlying DGP in sufficient detail.

[5] *Causal* DAGs differ from the broader class of DAGs in two key aspects: First, an arrow (→) between two variables, for example, $X \rightarrow Y$, always represents a causal relationship—meaning that a real-world intervention on X affects or at least has the potential to affect Y, whereas intervening on Y cannot affect X. Second, a causal DAG must include all common causes of any pair of variables in the graph. See the Appendix for more details.

require as rigorous a formalization of the DGP. For instance, rather than specifying the entire DGP, the Rubin Causal Model instead relies on a probabilistic model of the treatment assignment mechanism and the unconfoundedness assumption. This assumption requires a set of observed covariates that makes treatment assignment independent of the potential outcomes (Imbens & Rubin, 2015). However, the Rubin Causal Model does not specify how researchers should identify the covariate set that satisfies unconfoundedness—it only states that such a set of observed covariates is required. Without any formal guidance, researchers often control for as many baseline covariates as possible in the hope of satisfying the unconfoundedness assumption (Keller & Branson, 2024). But as mentioned earlier, adjusting for more variables does not necessarily decrease bias, it could also increase bias (Steiner & Kim, 2016; Wysocki et al., 2022). Covariate selection must therefore be defended on principled grounds, which means that assessing the unconfoundedness assumption from the Rubin Causal Model requires, in general, the very same qualitative background knowledge that causal graphs formalize. Thus, the argument that causal graphs cannot be used in practice because we do not have complete knowledge of the DGP applies equally and as forcefully to other causal frameworks such as the Rubin Causal Model. The advantage of causal graphs is that they force researchers to gather more information about the DGP and then explicitly formalize and defend their causal assumptions, rather than leaving them unstated, which then permits defending variable selection on principled grounds (Grosz et al., 2020).

There is no doubt that fully understanding the DGP and constructing a corresponding causal DAG is challenging. However, if we are to take causal inference from observational and experimental data seriously, we must develop a causal graph that is sufficiently credible for drawing valid causal conclusions. Pearl (2014, p. 463) highlights this aspect by stating that "[t]he purpose of the diagram is to provide an unambiguous description of the scientific context of a given application. While the application itself is usually shrouded in ambiguities and disagreements, the diagram represents a hypothetical consensus on what is plausible and important versus that which is deemed negligible or implausible." How, then, should researchers acquire the knowledge required to draw a credible causal graph to answer their research question? Researchers must carefully investigate (i) how subjects selected or got assigned

into treatment conditions (selection process), (ii) the causal determinants of the outcome (outcome-generating process), (iii) how these variables were defined, operationalized, and measured (construct validity), and (iv) how all of these variables are causally related to each other.

Gathering this information typically requires a combination of different, mostly qualitative methods. For instance, interviews or focus group discussions with study participants and administrators to understand (i) why participants selected or were assigned to different treatment conditions and (ii) what behaviors led to the observed outcome;[6] Literature reviews and subject-matter expert interviews to understand the conceptual definition of constructs and variables as well as their causal determinants and interrelations (Rodrigues et al., 2022); Examination of survey and measurement manuals to understand data collection procedures and construct validity; And, reviews of similar empirical studies to learn about causal interrelations between variables involved in the DGP. Collecting the causal input for valid causal inference involves effort and includes not only high-level theories of the phenomena under consideration but also attention to local, context-specific aspects of the research design that produced the data—including the study setting (participants, time, location), sample selection procedures, attrition and nonresponse, or instrumentation and measurement issues (e.g., order effects).

The second barrier is that the *process* of drawing valid causal graphs is difficult and the current literature does not provide thorough guidance on how to draw a causal graph systematically for practical applications. Even if we had full knowledge of the DGP, it remains challenging to reliably translate that knowledge into a causal graph without overlooking important aspects. Drawing causal graphs is a skill that must first be learned. This is in direct contrast to Pearl and Mackenzie (2018, p. 7) who claim that "these diagrams are extremely easy to draw." The lack of guidance and confusion about what causal graphs are good for can be attributed to the fact that the methodological literature almost exclusively focuses on "toy examples" and rarely provides more realistic examples. While toy examples are helpful for learning the graphical language and analysis, they neither convince researchers to use causal graphs

[6] For a discussion of qualitative methods such as process tracing and analytic narratives for causal inference with single cases see Maxwell (2012)—see also Cartwright (2022) and Humphreys & Jacobs (2023).

for their own applied research nor instruct them on how to implement them. Thus, it is not surprising that graphs are often found to be less useful—if not detrimental—for actual research practice (Bastías, 2024).

For instance, in comparing causal DAGs to the potential outcomes framework, Imbens (2020) states that the DAG literature lacks substantive empirical examples and instead focusses on the identification of causal effects in "toy" models. But if "the DAG literature has not shown much evidence of the benefits for empirical practice" (Imbens, 2020; p. 1132), it is not because they lack the potential to do so, but rather because methodologists have not yet provided applied researchers the practical guidance on how to use them in their own work. The lack of guidance is now being realized in the field, resulting in several articles offering some guidance for drawing causal graphs. However, most are rather short, lack comprehensive step-by-step instructions, have only a specific focus, or fail to address issues such as measurement error in observed variables or the importance of drawing both observed and unobserved confounders between any two variables already in the graph (Barnard-Mayers et al., 2022; Bastías, 2024; Digitale et al., 2022; Ferguson et al., 2020; Greenland, 2010; Huntington-Klein, 2022; Robins, 2001; Rodrigues et al., 2022; Tennant et al., 2021).[7]

The goal of this article is to move beyond toy examples and offer more comprehensive guidance on constructing credible causal graphs for applied settings in the social sciences, with the aim of promoting their use in applied research. To this end, we propose a procedure that begins with the causal effect of interest and incrementally adds variables and causal links (arrows) until a complete, causal graph is obtained. Unlike other approaches that assume that all the (un)observed variables involved in the DGP are known before drawing the graph (Ferguson et al., 2020; Rodrigues et al., 2022),[8] our method encourages researchers to identify and include relevant variables iteratively as they construct the graph. Before

[7] Huntington-Klein (2022) and Barnard-Mayers et al. (2022) have so far provided perhaps the most comprehensive, applied instructions for drawing causal graphs. Greenland (2010) offers an excellent critical perspective on issues that need to be considered and reflected in a causal graph. Howe et al. (2022) discuss causal graphs for disparity research. Rodrigues et al. (2022) highlight the importance of engaging with domain experts and propose ideas for structured workshops with experts to draw credible causal graphs. Hernán & Robins (2023) provide careful discussions about several aspects important for drawing DAGs (including issues like measurement error or variables lacking well-defined interventions).

[8] Given the set of all relevant variables, one starts by drawing a fully saturated graph in which all variables are connected, and then continues by directing or dropping edges according to subject-matter expertise.

outlining our stepwise procedure, we first introduce the setup of the hypothetical study that will serve as an example throughout the discussion.

Hypothetical Study: Online vs. In-Person CBT

We describe a study that aims to estimate the average causal effect of an online vs. in-person cognitive-behavioral therapy (CBT) on depression, which serves as the scenario for drawing a causal DAG. Once the completed causal graph is drawn, we briefly demonstrate how a graphical analysis helps researchers assess (i) whether the causal effect is identified for the observed data, (ii) which covariates to choose to adjust for confounding bias, and (iii) what can be done if the causal effect is not identified. These discussions will also demonstrate that graphs should ideally be used before data are collected to address potential identification issues in advance.

The participants in our hypothetical study are college students seeking treatment for depression at the university's counseling center. After enrolling in the study, the students fill out an intake questionnaire that collects information about their pre-treatment depression score using the Beck Depression Inventory (P), their socio-demographic characteristics which comprise their race, gender, and age (S^*), and their history of medication and substance use before the treatment (M_0^*). Then, the students' brain scans are recorded on-site to measure their cognitive-affective mechanism (CAM) before treatment (C_0^*). The CAM refers to the neural brain processes that generate the symptoms of depression according to Beck's cognitive model (e.g., rumination, dysfunctional schemas, and biased attention, processing, and memory towards negative emotional stimuli; Disner et al., 2011). After all measurements have been taken, researchers then assign participants to the in-person CBT treatment ($T = 0$) or the online CBT treatment ($T = 1$) based on an informal assessment of the participants' pre-treatment depression score (P), socio-demographic characteristics (S^*), and pre-treatment CAM (C_0^*).

The CBT treatment is administered twice a week over three months. Immediately after the final CBT session, the students complete an exit survey that measures their post-treatment depression score using the Beck Depression Inventory (Y) and their medication and substance use after treatment (M_1^*). Thus, we have seven observed variables in total: T, P, Y, S^*, C_0^*, M_0^*, and M_1^*. But note that S^* is a set of

three variables (race, gender, age) and M_0^*, and M_1^* are sets of at least two variables (medication and substance use). Variables can be grouped into sets if they serve a similar function within the DGP, even if their effects operate in opposite directions. Grouping also enhances the readability of the graph by reducing visual complexity. Variables subscripted with 0 are measured at the baseline, that is, before the onset of the treatment (including P). Variables with subscript 1 are measured after the treatment (including Y). All variables with the same subscript are assumed to be measured on the same day. A star superscript (*) indicates a fallible measure of an underlying construct. For instance, C_0^* is a noisy brain scan of the actual brain activity C_0, and race, gender, and age (S^*) do not fully capture all socio-demographic variables of relevance (such as parents' unobserved education and income which likely confound some of the variables).

Since researchers partially control several aspects of the study, such as the treatment conditions, the assignment process, and the measurement time points, we refer to it as a quasi-experiment. This allows us to later distinguish it from an observational study, which differs only in that treatment selection is based on self-selection rather than researcher assignment. Using the quasi-experiment as an example, the following sections will demonstrate the steps of drawing a causal graph that reflects the DGP related to the treatment effect of interest. To obtain a credible causal graph, we consulted with subject-matter experts such as mental health practitioners, researchers, and psychometricians. The graph we draw here attempts to synthesize their expertise. Because our example is hypothetical, we did not need to consult the (hypothetical) study's researchers or participants to find out how participants were assigned to different treatment groups, but in actual practice, the assignment mechanism must be carefully understood. Ideally, the final causal graph represents a consensus among researchers and experts about the relevant variables and their causal relations in the DGP (Pearl, 2014).

However, consensus might not be reached, as researchers and experts may disagree with which graph adequately represents the study-specific DGP. For example, we will see that our causal graph reflects the perspective that depression is fundamentally driven and explained by brain processes, while also acknowledging that current brain imaging techniques are noisy and capture these processes with

measurement error. Researchers sympathetic to the more traditional, latent variable view of depression may draw the graph differently, which could lead to different identification results. However, this reflects the strength of this framework. By providing a language to formalize causal assumptions explicitly, causal graphs demand that researchers justify the credibility of their assumptions (such as whether depression is reducible to neural brain processes), which otherwise gets taken for granted (Grosz et al., 2020). Although we discussed our causal graph with experts and psychometricians, we do not intend to claim here that the causal graph we propose is the *only* possible causal DAG that can be constructed for our hypothetical study. Instead, our goal is simply to provide concrete steps to structure the *process* of drawing credible causal DAGs, which can then facilitate discussions about the graph's accuracy and credibility.

Steps to Drawing Causal Graphs

Overview

From a technical perspective, causal graphs seem easy to draw. Given a treatment variable (T) and an outcome (Y), one "only" needs to add all common causes (confounders) for each pair of variables in the graph. That is, after drawing the causal connection(s) between the treatment T and the outcome Y, as well as the variables that affect T and Y, one starts adding common causes for each pair of variables until no further common causes of any variable pair can be identified. However, "variables" refer to (i) both observed and unobserved constructs involved in the DGP, (ii) variables related to the study design like blocks, strata, eligibility or sample selection criteria, and other data issues (e.g., nonresponse, attrition), and (iii) any fallible measures of DGP-relevant constructs (an aspect rarely covered in the literature; but see Greenland, 2020, Hernán & Robins, 2023, or Steiner & Kim, 2016). This means that a causal graph has to describe all the processes that produce the *observed* data of the study, which can easily go awry if done unsystematically.

To draw causal graphs more systematically, we describe a procedure of six partially iterative steps, which are summarized in Table 1 along with questions to consider at each step. The six steps consist of drawing the following aspects of a DGP:

(1) *Causal Process*: the causal path(s) from the treatment to the outcome of interest,

(2) *Selection Process*: the causal process of how individuals get assigned or select into treatment conditions,

(3) *Outcome-Generating Process*: the causal determinants of the outcome,

(4) *Pairwise Interrelations*: the causal and confounding relations among variables in the graph,

(5) *Additional Observed Variables*: any remaining and relevant observed variables in the data set, and

(6) *Iterate Steps 4 & 5*: add (new) pairwise causal relations arising from steps 4 and 5.

Step 1: Causal Process

The first step is to draw all causal paths of interest that connect the treatment indicator and the outcome. Causal paths include both the direct effect from the treatment to the outcome and any indirect effects via mediators of interest. If mediators are not of interest, only the total effect $T \rightarrow Y$ needs to be drawn, which then captures all mediated and unmediated effects.

In our study, the treatment variable indicates the in-person CBT ($T = 0$) vs. the online CBT ($T = 1$), and the outcome is the post-treatment depression score (Y). We believe that the effect of the CBT treatment on depression is entirely mediated by the latent CAM at the posttest (C_1), which we assume is not measured in this study. In other words, any difference in the treatment efficacy between online and in-person CBT is due entirely to differential changes in the CAM.[9] Therefore, we begin by drawing the treatment variable T and the outcome Y and connect them with arrows via C_1: $T \rightarrow C_1 \rightarrow Y$. This causal path consists of two links, $T \rightarrow C_1$ and $C_1 \rightarrow Y$. For this study, there is only one *causal* path between T and Y (see Figure 1).

The research question for this study investigates whether the path $T \rightarrow C_1 \rightarrow Y$ transmits causal association and how large the effect is. Given that our research question is not about mediation processes and that the post-treatment CAM, C_1, is unobserved, we could choose to omit this variable and simply draw a direct arrow from T to Y instead: $T \rightarrow Y$. In most cases where the average causal effect is of

[9] Some researchers might argue that the CAM does not fully capture the data-generating depression construct and that the intervention impacts the depression outcome via other mediators, for instance, via a more traditional psychometric depression construct or via other cognitive brain activities. To accommodate such claims, we could either broaden the scope of both C_0 and C_1 to cover the other mediating processes or explicitly drawn them.

interest, drawing only the direct causal path $T \rightarrow Y$ represents the total effect of T on Y. However, by drawing the unmeasured CAM, we explicitly postulate the mechanism through which we believe the treatment exerts its impact, which not only helps with later steps of drawing the graph but also emphasizes the idea that we are not merely drawing measured data, but the actual DGP, regardless of whether the variables are measured.

Step 2: Selection Process

The second step is to draw the selection process, which describes how participants were assigned to or self-selected into different treatment conditions. In experimental studies, researchers control how participants get assigned to treatment conditions, whereas in observational studies, participants self-select or get assigned by third parties like administrators into treatment conditions.[10]

In this study, researchers assigned the participants to either the in-person CBT ($T = 0$) or the online CBT ($T = 1$) based on three measured variables: their pre-treatment CAM (C_0^*), pre-treatment depression score (P), and socio-demographic variables (S^*: race, gender and age). Specifically, researchers deliberately decided that participants at greater risk—those with high depression pre-treatment scores, marginalized group membership, and greater markers of depression in their CAM before treatment—should be assigned to the in-person CBT rather than the online CBT.[11] Therefore, we add the three variables, C_0^*, P, S^*, and draw corresponding arrows from these variables to the treatment indicator, $C_0^* \rightarrow T$, $P \rightarrow T$, and $S^* \rightarrow T$ (see Figure 2). Importantly, we draw the variables C_0^* and S^* with a star superscript because they are fallible measures of underlying constructs—as is the pretest depression measure P. Brain scans are noisy (i.e., unreliable) due to current limitations in brain imaging techniques,

[10] This step in drawing a causal graph refers to the *causal* selection process and, thus, is in general not identical to the "assignment mechanism" of the Rubin Causal Model (Imbens & Rubin, 2015, 2018). In the Rubin Causal Model, the assignment mechanism is a probabilistic rather than a causal model of treatment assignment or selection (the assignment probabilities are a function of observed baseline variables and the unobserved potential outcomes which are post-treatment variables—unobserved baseline variables cannot be part of the assignment mechanism). However, in special cases like randomized (block) designs or for quasi-experimental designs with known assignment variables, the probabilistic model is identical to the causal model.

[11] Note that there are no explicit cutoff values in place here as there are in regression discontinuity designs. Thus, researchers informally assigned students to the two CBT conditions based on subjective assessments of brain scans, depression pretest scores, and three socio-demographic variables. Other individual factors may have influenced the assignment decision, but these are assumed to be idiosyncratic and non-systematic.

and socio-demographic variables may not be perfectly measured as both race and gender are multidimensional constructs, and single measures of race and gender fail to distinguish between self-identified versus perceived race and gender. Moreover, other relevant socio-demographic factors, such as parents' education and income, are not measured at all (see Step 3). Thus, S^* is an incomplete set of socio-demographic measure. Since assignment to treatment is actually based on the unreliable measures, we must link those measures rather than the underlying true variables or constructs to the treatment indicator.

The example highlights how the selection process is straightforward to draw when the variables that determine treatment selection are both *known* and *measured* as part of the study design. The fact that the selection mechanism is known and measured is a reason why causal inferences drawn from randomized and quasi-experiments are generally more credible than those from purely observational studies. When participants are instead free to choose their preferred treatment condition, determining the selection mechanism becomes more challenging. Nonetheless, researchers must identify all the relevant variables that influence participants' treatment choice, which requires greater subject-matter knowledge about participant behavior—potentially informed by participant interviews or other qualitative methods.

Step 3: Outcome-Generating Process

The third step requires drawing the outcome-generating process, which describes how the outcome values are determined (Figure 3). In our study, the outcome of interest is the depression post-treatment score (Y) with the latent intermediate post-treatment CAM (C_1). First, we believe that the initial CAM before treatment (C_0) affects the CAM after treatment (C_1) because of natural growth or stability over time, $C_0 \rightarrow C_1$. That is, the post-treatment CAM is assumed to depend on the baseline CAM; a participant who starts the study with a higher level of depression will likely end the study with a higher level than someone who starts with a lower level. Second, we believe that socio-demographic student characteristics (S; race, gender, age, parents' education and age, the size and quality of family and peer networks) affect the post-treatment CAM (C_1), $S \rightarrow C_1$. For example, members of marginalized groups experience more discrimination such as racism and sexism, and the resulting minority stress can lead to or worsen depression. Additionally, younger college students are generally more at risk for depression to the

challenges of a major life transition and limited experience with coping strategies. Furthermore, we also believe that the socio-demographic variables (S) affect the outcome (Y) directly, $S \rightarrow Y$, because socio-demographic factors such as race, gender, or parents' socio-economic status may influence responses to the questionnaire through differential item functioning, even after controlling for its indirect effect through the CAM (C_1), $S \rightarrow C_1 \rightarrow Y$.

We should highlight here that it is the latent pre-treatment CAM (C_0) and the true socio-demographic variables (S) that affect the mediator (C_1) and outcome (Y), not their unreliably measured counterparts (C_0^* and S^*). Generally speaking, it is important to distinguish clearly between the latent variables and their corresponding (unreliable) measures because both may play different roles in the DGP regarding treatment assignment and the outcome-generating process. While the unreliable measures may directly determine treatment assignment (as in our case), they often do not directly affect the outcome. Ignoring this distinction and drawing only the (fallible) observed variables (e.g., $C_0^* \rightarrow Y$ and $S^* \rightarrow Y$) can lead to false claims derived from the graph about whether the causal effect is identified with the data and result in misleading causal inference. For example, drawing these causal relations using the fallible measures, $C_0^* \rightarrow Y$ and $S^* \rightarrow Y$, would imply that if the brain scan data were incorrectly preprocessed or if socio-demographic variables were misreported on the intake forms, then these measurement errors would directly alter the student's post-treatment depression outcome Y. But this is implausible for our example.[12] Nonetheless, the measured CAM (C_0^*) and socio-demographic variables (S^*) are associated with the post-treatment depression along noncausal paths as will become clear later in step 4 ($C_0^* \leftarrow C_0 \rightarrow Y$ and $S^* \leftarrow S \rightarrow Y$). However, if the brain scans and socio-demographic variables were measured with perfect

[12] In other situations, it may be possible for fallible measures like P or C_0^* to affect the outcome C_1 or Y, particularly so if students were to know these measures and then change their behavior. For example, students who find out that their brain scans show significant signs of depression or that their pre-treatment depression score is extremely high may become demotivated, which can worsen depressive cycles. But we assume that this is not the case in our study because participants neither learned about their brain scans nor their pre-treatment depression scores. To assess whether a fallible measure has a direct impact on other variables it is helpful to consider whether a real-world intervention on the fallible measure instead of the underlying true construct (e.g., by intentionally recording or reporting false values) would actually affect other variables in the DGP.

reliability with respect to the true variables involved in the DGP, then $C_0^* = C_0$ and $S^* = S$, and the distinction between the latent and measured variables would disappear and only C_0 and S would need to be drawn.

Furthermore, we add three additional variables that affect depression via the post-treatment CAM (C_1): genetic predispositions towards depression (G), physical health after treatment (H_1), and medication and substance use after treatment (M_1). Depression is partially heritable as genes can predispose individuals towards developing depression, $G \rightarrow C_1$, better physical health serves as a protective factor against developing depression, $H_1 \rightarrow C_1$, and psychotropic medication like antidepressants can decrease depression whereas substance abuse increases the risk for depression, $M_1 \rightarrow C_1$.

If mediating processes are not of interest, it would be sufficient to use *baseline* health status (H_0) and medication and substance use (M_0), rather than their values at the time of outcome measurement, and connect them directly to CAM: $H_0 \rightarrow C_1$ and $M_0 \rightarrow C_1$, respectively. However, we prefer to represent the corresponding constructs at the time point of the outcome measure because they more explicitly reflect the causal, time-varying processes in greater detail, and thus in a more understandable way.[13] The corresponding baseline measures will enter the graph in step 4.

Step 4: Pairwise Interrelations

The fourth step is to examine each pair of variables drawn so far and decide if there should be a (i) direct causal relation, (ii) confounding relation, or (iii) no relation between them (Figure 4). This step is arguably the most difficult because of the potentially large number of pairs to consider for most realistic graphs. For a graph with n variables, there are $n(n\text{-}1)/2$ total pairs of variables to consider. First, for each pair of variables, determine whether there is a direct causal effect that must be included above and beyond any indirect causal relations already captured by the graph. If so, the direct *causal relation* between the two variables must be represented by an arrow in the causal direction. Then, again for each

[13] This is particularly relevant when causal DAGs are used to assess the plausibility of the common trend assumption in difference-in-differences designs or to represent time-varying mediation processes and dynamic treatment regimes.

pair of variables, check whether they need to be connected by a common cause, that is, a variable not yet in the graph that jointly affects the two variables.[14] If so, add the common cause variable, even if unobserved, and draw arrows pointing into the two variables. Treating the common causes as a separate step explicitly highlights that a "causal" DAG is obtained only if it includes all common causes of each variable pair.

Our graph includes 11 variables so far (Figure 3): G, H_1, C_0, C_0^*, P, M_1, S, S^*, T, C_1, Y. Thus, we have 55 pairs of variables to consider. To carry out this step systematically, the supplemental materials provide R code using the DAGitty package (Textor et al., 2011, 2016) that lists every pair of variables in the graph that have not (yet) been connected by an arrow.

Step 4(a): Presence or Absence of Direct Causal Relations. In this step, we determine for each pair of variables whether they should be connected by an arrow that represents a direct effect. To illustrate some general principles, we discuss two examples here where an arrow for a direct effect must be drawn and two examples where we argue that no arrow needs to be drawn. When deciding whether to draw a direct arrow between two variables, always consider their temporal order. As a general rule, only variables that occur earlier in time can causally influence those that occur later. Temporal order thus constrains the possible direction of any presumed causal effect.[15]

Direct Effects $C_0 \rightarrow C_0^*$ and $S \rightarrow S^*$. The latent pre-treatment CAM (C_0) should be directly connected to the measured noisy brain scans at the baseline (C_0^*), and the true socio-demographic variables (S) should be directly connected to the fallible measures of the socio-demographic variables

[14] For two variables X and Z, the variable C is a common cause of X and Z if C directly affects both variables: $X \leftarrow C \rightarrow Z$. A common cause induces a *confounding relation* between the two variables.

[15] Three remarks are in place here. First, reciprocal causation where two variables affect each another seemingly simultaneously are discussed in step 6. Second, the measurement time point of variables does not always determine the causal ordering—for instance, when measurements are based on retrospective questions about the past (but reliability issues might jeopardize causal identification here). And third, any unobserved variables in the graph must refer to a clear time point so that the causal ordering of all variables in the graph is possible.

(S^*): $C_0 \rightarrow C_0^*$ and $S \rightarrow S^*$. As a general rule, a latent variable should always be connected to its measured counterpart. These two examples are aligned with reflective measurement models in psychometrics.[16]

Direct Effects $S \rightarrow C_0$ and $G \rightarrow C_0$. Many socio-demographic variables (S) are implicated in the development and maintenance of depression (C_0). For instance, individuals from marginalized genders and races are more likely to be discriminated against, and the stressors from these variables can contribute towards depression. Therefore, we need to add an arrow to represent the effect: $S \rightarrow C_0$. Note that the justification for this direct effect is exactly the same as for the arrow from $S \rightarrow C_1$ from step 3 (outcome-generating process). Thus, variables with direct effects on a variable at one time point (e.g., C_1) often have an effect on the same variable at another time point (e.g., C_0). The same principle applies to genetic predisposition (G), which already has an effect on C_1, so we add the arrow connecting $G \rightarrow C_0$.

Missing Arrow $G \nrightarrow P$. We argue that genetic predisposition (G) does not directly cause the pre-treatment Beck Depression Inventory score (P) because there is already an indirect effect from G on P via the latent pre-treatment CAM (C_0): $G \rightarrow C_0 \rightarrow P$. Thus, we claim that there is no plausible causal mechanism other than via C_0 of how genetic predisposition (G) affects the pre-treatment depression score (P). This illustrates the importance of considering whether there are already indirect effects connecting the two variables because the direct effect must reflect a causal process beyond what those indirect effects already represent.

Missing Arrows $P \nrightarrow C_0^*$ and $P \nleftarrow C_0^*$. There is no direct impact of P on C_0^* because we assume that measuring the Beck Depression Inventory (P) does not affect depression itself (via the latent CAM) and therefore should not influence the brain scan measures of depression either. We also believe that the baseline CAM brain scans (C_0^*) do not directly cause the baseline depression score (P). In this study, participants first fill out the Beck Depression Inventory and subsequently have their brains scanned. Due

[16] It is also possible that a formative model fits the DGP better, particularly so if the observed items rather than the formative constructs play the predominant role in the DGP (this issues require careful consideration, but are beyond the scope of this introduction; Vanderweele, 2022).

to the temporal ordering of measurements, the brain scans (C_0^*) cannot directly influence the depression score (P).

The direct effects between all pairs of variables are shown in Figure 4. Direct effects and missing arrows are justified by different processes, such as temporal ordering (e.g., causes must come before effects), measurement theory about the relationship between latent and measured variables (e.g., measured variables are often unreliable proxies of the true latent variable), and theory about causal mechanisms (e.g., CBT is thought to affect depression solely through the CAM).

It is important to note that the *omission* of arrows in a causal graph is both more stringent and more consequential for causal inference than their *inclusion* (Greenland, 2010). This is because missing arrows encode the assumption that the direct effect is exactly zero for most if not all participants. The more missing arrows a DAG includes, the more likely the causal effect of interest can be identified because the absence of certain paths can reduce the number of variables that need to be adjusted for. However, if omitting some arrows is unwarranted, then the model may misidentify causal effects and result in biased causal estimates. Thus, each missing arrow must be thoroughly defended by arguing that such direct effects are either logically impossible, substantively implausible, or negligibly small (i.e., close to zero for all subjects or relevant only for a negligibly small subgroup). Whenever possible, such claims should be supported with (qualitative) empirical evidence or citations from the relevant literature.

When in doubt between including or omitting an arrow, the more cautious approach is to always draw the arrow. Adding an arrow when no direct effect exists may unnecessarily complicate causal identification (e.g., requiring more variables to adjust for confounding bias), but it protects against false-positive identification claims. Ultimately, the process of examining each pair of variables is a useful exercise that encourages researchers to systematically construct a defensible theory about the presumed DGP and to thoroughly justify the absence of direct effects. When evaluating causal graphs, attention should focus on what is *not* shown—the omitted arrows and variables—rather than what is explicitly included.

Step 4(b): Confounding Relations. Completing Step 4(a) guarantees that the graph correctly represents the common causes of each pair of variables, but only for the variables included so far in the graph. Two variables might still be associated due to a common cause not yet shown in the graph. Thus, we need to examine whether each pair of variables needs to be connected by common causes that are not yet in the graph. This entails adding new common cause variables with arrows pointing into both variables in the pair. For a given variable pair, more than one common cause might need to be added and, moreover, all the common causes might be difficult to list. In such cases, researchers might add an unspecified variable that captures all presumed common causes. These common causes may be designated with labels U_1 , U_2, U_3, etc. since they are often unobserved (U).

Physical health after treatment (H_1) and the baseline CAM (C_0) are both assumed to be affected by pre-treatment physical health (H_0). Therefore, we add H_0 as a variable and draw the arrows $H_0 \rightarrow H_1$ and $H_0 \rightarrow C_0$ to represent the common cause structure $C_0 \leftarrow H_0 \rightarrow H_1$. Second, the pre-treatment medication and substance use (M_0) is a common cause of baseline CAM (C_0) and post-treatment CAM (C_1), $C_0 \leftarrow M_0 \rightarrow C_1$, but also of post-treatment CAM (C_1) and mediation or substance use (M_1), $C_1 \leftarrow M_0 \rightarrow M_1$. Therefore, we add M_0 as a new variable and add the arrows: $M_0 \rightarrow C_0$, $M_0 \rightarrow C_1$, and $M_0 \rightarrow M_1$. The common causes are shown on Figure 5.

Step 5: Additional Observed Variables

The fifth step is to add any additional variables that are observed or planned to be measured that are related to the variables already drawn in the graph. These may include fallible measures of data-generating constructs, exogenous variables (i.e., idiosyncratic causes of single variables), or variables on the causal pathway between pairs of variables (i.e., intermediate variables). This step helps clearly distinguish between the true, potentially latent variables involved in the DGP, and observed proxies or fallible measures of these variables.

In this study, the two remaining observed variables are baseline medication and substance use (M_0^*) and the corresponding post-treatment measure (M_1^*), which are fallible measures of the true medication and substance use at the baseline (M_0) and post-treatment (M_1), respectively. The distinction

between measured versus latent medication and substance use is relevant because participants often inaccurately report their medication and substance use, intentionally or accidentally, and it is their true medication use that has an effect on their depression score, and not the fallibly measured one. Therefore, we add two arrows: $M_0 \rightarrow M_0^*$, $M_1 \rightarrow M_1^*$. The observed variables are added in Figure 6.

Step 6: Iterate Steps 4 & 5

The sixth and final step is to repeat steps 4(a) and 4(b) and check for all pairs containing at least one new variable from step 4(b) or step 5 whether they should be connected by a direct causal relation or a confounding relation. Since we added four new variables to our graph—M_0^*, M_0, M_1^*, H_0—we have to assess the causal and confounding relations among 50 new pairs.[17]

Here, we draw attention to the physical health (H_0, H_1) and CAM (C_0, C_1) variables because we believe they affect one another over time but not simultaneously. Specifically, we believe that past physical health (H_0) affects current CAM (C_1), because poor physical health may lead to greater depression. But we also believe that past CAM (C_0) affects current physical health (H_1), because depression may lead to poorer physical health. The way we capture this bi-directional relationship is by explicitly showing how the relation between health and CAM unfolds over time: baseline physical health affects the post-treatment CAM, $C_0 \rightarrow H_1$, and the baseline CAM impact post-treatment physical health: $H_0 \rightarrow C_1$. However, here we also draw the direct relation $H_1 \rightarrow C_1$ seemingly suggesting an immediate causal effect. However, here we mean that the past health condition between the time period 0 and 1 has an impact on CAM at the time point of measurement (at time 1). This helps us avoid overloading the graph with too many variables, but as this example highlights, causal graphs may look different if the meaning of variables and their temporal reference changes.

The mutual influence of physical health and the CAM is also present before the measurement at baseline, that is, pre-baseline physical health (H_{-1}) and CAM (C_{-1}) impact baseline physical health (H_0) and CAM (C_0). Specifically, H_{-1} affects C_0, and C_{-1} affects H_0 for the same reasons mentioned earlier.

[17] (11+4)(11+4-1)/2 - (11)(11-1)/2 = 50

Furthermore, the pre-study physical health (H_{-1}) affects the pre-treatment physical health (H_0), and the pre-study CAM (C_{-1}) affects the pre-treatment CAM (C_0). Therefore, we add H_{-1} and C_{-1} as new variables and add the following links: $H_{-1} \rightarrow H_0$, $H_{-1} \rightarrow C_0$, $C_{-1} \rightarrow H_0$, and $C_{-1} \rightarrow C_0$. The same argument would now apply to H_{-1} and C_{-1} and demand the inclusion of C_{-2} and H_{-2}, and so on. To avoid an infinite regress here, we define the pre-study variables C_{-1} and H_{-1} as covering the entire history of CAM and physical health but connect C_{-1} and H_{-1} with an unspecified common cause U, $C_{-1} \leftarrow U \rightarrow H_{-1}$. Using time-indexed variables allows us to unfold the causal process over time and thereby avoid issues related to reverse causation and cycles, which are by definition not permitted in *directed acyclic* graphs.

The Final Causal Graph

Figure 9 shows the final causal DAG after completing all six steps. By claiming that the DAG is a *causal* graph, we are declaring our belief that the graph includes all common causes in the DGP—which was addressed in steps 4 and 6. This implies that any omitted causes of each variable must be variable-specific idiosyncratic causes that are mutually independent of each other. Although the idiosyncratic causes (often referred to as the error terms) are typically not drawn in the graph, it is important to always keep them in mind.[18] First, they define the time-specific context in which the average causal effect is evaluated. For example, the causes subsumed in the error term of the outcome (or any mediators) are often effect moderators. Thus, for different subpopulations or contexts defined by these idiosyncratic causes we would obtain different average causal effects. In our graph, the population of interest was depressed college students. If we believed that the average causal effect would be different if we studied depressed employees of the same age, the effect heterogeneity due to the differences in the setting and population—college vs. workforce—would be captured by the idiosyncratic cause on the depression outcome. Second, idiosyncratic causes of treatment selection could serve as an instrumental variable and allow for the identification of causal effects in the presence of unobserved confounding (Angrist et al.,

[18] Given a *causal* graph, all error terms (idiosyncratic causes) are mutually independent. Thus, when linking the causal graph to probability theory (e.g., via a structural causal model), the joint distribution of the error terms is simply the product of the error terms' marginal distributions.

1996; Sheehan et al., 2024). Thus, sometimes it is worth drawing them, for instance, when a pandemic hits in the middle of a study, or if a specific variable qualifies as an IV for treatment receipt.

Graphical Analysis

After the qualitative background knowledge about the DGP has been formalized into a causal DAG, the graph can be used to (i) derive testable independence relations between observed variables (and adjust the causal graph if needed), (ii) check whether the observed data allow for the identification of the average causal effect, (iii) determine which adjustment sets of observed covariates remove the entire confounding bias, and (iv), in case the average causal effect is not identified, determine all open noncausal paths and assess, together with additional knowledge about the DGP, the direction and magnitude of the remaining confounding bias. Importantly, all these aspects can be addressed directly and automatically by a graphical analysis of the DAG without collecting any data. Software packages like DAGitty (Textor et al., 2011, 2016) can be used to implement the analyses. We briefly highlight some results of a graphical analysis (for a brief introduction, see the Appendix; for the underlying theory see Elwert, 2013; Pearl, 2009; Pearl et al., 2016; Peters et al., 2017; Spirtes et al., 2000; Steiner et al., 2023). All results of the graphical analyses can be reproduced with the R script in the supplemental materials.

The graphical analysis reveals that without conditioning on any variables, Figure 9 includes 1 causal path, 296 open confounder paths, and 11,148 closed collider paths connecting the CBT treatment (T) variable and the depression outcome (Y). Assuming that the causal graph correctly reflects the underlying DGP, any observed association between the treatment and outcome variable can only be due to the associations transmitted along the open causal and the 296 confounder paths. If this were not the case, the causal graph would be incorrect, suggesting that some common causes are still missing. The graph has 2 empirically testable conditional independencies—$M_0^* \perp\!\!\!\perp T \mid C_0^*, P, S^*$ and $M_1^* \perp\!\!\!\perp T \mid C_0^*, P, S^*$—which can be assessed nonparametrically using Kernel-based conditional independence tests (e.g., Zhang et al., 2012).

Next, we check whether the observed covariates allow us to identify the average causal effect and which covariates sets researchers may use to remove the entire confounding bias. The minimal adjustment

set of observed variables that blocks all confounder paths while keeping all collider paths closed is $\{C_0^*, P, S^*\}$, which suggests that the causal effect is identified.[19] The observed adjustment set can then be used to estimate the causal effect—for instance, through inverse-probability-of-treatment weighting or regression-based covariate adjustment (e.g., Imbens & Rubin, 2015; Schafer & Kang, 2008).

However, the effect is also identified if we add one or both of the remaining observed variables, M_0^* and M_1^*, to the adjustment set. For instance, conditioning on $\{C_0^*, P, S^*, M_0^*, M_1^*\}$ would remove the entire confounding bias even though the set includes a post-treatment variable (M_1^*). The Rubin Causal Model generally recommends using only pre-treatment variables as control variables, since this rules out the possibility of controlling for a mediator (Imbens & Rubin, 2015, p. 265). However, the causal graph suggests that conditioning on the post-treatment variable M_1^* does not cause any problems.

Causal Graphs of Other Data-Generating Processes

In the previous example, we showed that the causal effect is identified if we condition on the three variables that make up the assignment mechanism the researchers set up by design $\{C_0^*, P, S^*\}$. Here, we consider two alternative scenarios that are commonly found in applied research settings, which have different considerations when identifying the causal effect: the observational study with self-selection and the randomized experiment. Though the research question remains the same as before (evaluating the average causal effect of online vs. in-person CBT) we discuss these alternative scenarios to demonstrate that causal graphs are context-dependent and vary with the research design and specific threats to validity. We also highlight that causal graphs are useful tools even in the context of randomized experiments.

Observational Study with Self-selection

[19] Note that this is the same set of variables involved in the assignment mechanism discussed in Step 2. Proponents of the potential outcomes framework would also agree that conditioning on these variables meets the unconfoundedness assumption. In this specific setting, where the variables that determine the selection process are known and observed, one would not need to draw the entire causal graph because the selection-relevant variables (i.e., the parental variables of treatment T) automatically block all noncausal backdoor paths.

In an observational study, the researchers do not control the assignment of study participants into the treatment conditions. Instead, participants self-select into either the in-person or online CBT where the treatment choice may depend on a number of different variables. Causal identification is more challenging in observational studies because the variables involved in the selection process are neither entirely known nor fully observed.

The causal DAG in Figure 10 shows a scenario where participants with a more depressive CAM (C_0) and high depression pretest measure (P) are more likely to choose the in-person CBT, and participants with more marginalized races and genders or from households with low socio-economic status (S) are more likely to choose the online version. Thus, the treatment selection mechanism is now represented by three arrows pointing into T: $C_0 \rightarrow T$, $P \rightarrow T$, and $S \rightarrow T$. Unlike the original quasi-experiment example, the treatment selection is based on two latent variables (C_0, S) and only one observed variable (P).

Since treatment selection is no longer based entirely on observed variables, we briefly discuss issues related to selection on unobserved variables or constructs in more detail. By drawing $C_0 \rightarrow T$, we assume that the baseline CAM causally affects some students' choice of CBT version. However, for this causal link to be meaningful, we must be able to conceive of a well-defined intervention on the CAM that changes it from a high-depression to a lower-depression state. While such a change might, in principle, be achieved by targeted, physical stimulations of certain brain areas, this remains a vague notion of an intervention, making the existence of a "true" causal relationship between CAM and treatment selection questionable (see Hernán and Robins, 2023, for a discussion of corresponding "noncausal" DAGs). The same concern applies to the causal relationship between CAM and the depression outcome at posttest, $C_1 \rightarrow Y$. The implication of a CAM construct without a well-defined intervention is that it becomes unclear which CAM-related measures would need to be taken to block any corresponding noncausal paths, as well as whether the measures obtained from brain scans (C_1^*) validly capture the construct involved in the DGP.

Similar challenges arise for the selection-relevant constructs of sex/gender and race. Here, too, it is unclear what a meaningful intervention would entail, since these constructs may refer to distinct aspects or signals. For instance, with regard to sex/gender, one may need to distinguish between biological sex, self-identified gender, and perceived gender (by others, such as peers). While self-identified gender and biological sex may influence CBT selection, perceived gender may particularly affect a student's CAM and depression level (assuming that the gender perceived by others affects a student's social interactions and mental conditions). Thus, it is conceptually useful to distinguish these aspects and represent them separately in the DAG because it facilitates the conception of concrete interventions on each aspect (Weinberger, 2023) and because it provides a stronger warrant for interpreting the arrows causally. It also guides researchers in deciding which variables or aspects of constructs to measure. However, in our current graph, these distinctions are not made explicit; They are subsumed under the variable set of socio-demographic characteristics (S). When deciding which variables to measure or when assessing potential residual bias, these distinctions must nonetheless be kept in mind.

The graphical analysis suggests that this selection process induces confounding associations along 296 open confounding paths, just as the quasi-experimental setting in Figure 9. However, unlike the quasi-experiment, conditioning on all measured baseline variables, $\{C_0^*, S^*, P, M_0^*\}$, is unable to close all confounding paths and, in addition, opens up collider paths: 148 open confounding paths remain open and 836 collider paths are unblocked which now transmit spurious collider association between T and Y. This example highlights the impact of measurement error on the identification and estimation of causal effects and why it is important to distinguish between the two when drawing causal graphs. If there were no measurement error, $C_0 = C_0^*$, and $S = S^*$, then conditioning on the measured variables would fully block the confounding paths and not open any collider paths.

However, given that $C_0^*$and S^* are fallible measures of C_0 and S, only the confounder paths via P are fully blocked while those via C_0 and S (but not traversing via P) remain open and still transmit confounding association. Inspecting the open confounder paths helps in assessing the direction and magnitude of the bias. Here we briefly discuss the main principles of such an assessment using only three

of the 148 open confounder paths: $T \leftarrow S \rightarrow Y$, $T \leftarrow S \rightarrow C_1 \rightarrow Y$ and $T \leftarrow C_0 \rightarrow C_1 \rightarrow Y$. First, given that the unobserved S and C_0 directly affect both T and Y or C_1, they have a high potential to significantly confound the causal relation, particularly if the impact of S and C_0 on both T and Y is large.[20] However, since all observed variables $\{C_0^*, S^*, P, M_0^*\}$ are (close) proxies of both S and C_0, they block the confounder paths to the extent of their squared correlation with the unobserved variables (this simple rule holds under linearity; Steiner & Kim, 2016). If measurement error in C_0^* and S^* is low (say, their reliability is at least .95), then the three paths are almost fully blocked, with only minimal residual confounding "leaking" through. The same logic extends to all other open confounder paths connecting T and Y, particularly longer ones that traverse via three or more variables: such paths only need to contain one or two weak causal links to transmit little confounding association to begin with[21], and their length provides multiple opportunities for proxies of variables on the path to partially block confounding.

The same arguments can be made for the open collider paths. Almost all the collider paths are long and transmit weak associations to begin with—unless all links on the path are strong (i.e., the respective variables of each link have a nearly perfect, direct causal relation). Moreover, the weak associations are further mitigated since conditioning on the observed variables $\{C_0^*, S^*, P, M_0^*\}$ partially blocks the collider paths at multiple variables on the path. Consider two short, open collider paths: $T \leftarrow C_0 \rightarrow P \leftarrow S \rightarrow Y$ and $T \leftarrow S \rightarrow C_0 \leftarrow G \rightarrow C_1 \rightarrow Y$. On the first path, P is the collider that induces collider association (after conditioning on P). Due to the length of the path and if S or C_0 has only a weak impact on at least one of their descending variables (P, Y, T), the collider association is small even before

[20] The relative magnitude of the confounding transmitted along these paths depends on the strength of each causal link. For instance, the path $T \leftarrow C_0 \rightarrow C_1 \rightarrow Y$ induces substantial confounding if C_0 strongly influence the CBT choice, because the link $C_0 \rightarrow C_1$ is presumably very strong and C_1 almost surely has a strong effect on the depression measurement Y. Assuming that the CAM is positively linked to both the selection of the online CBT and the depression outcome, this path transmits a positive confounding association (VanderWeele & Robins, 2010). A similar assessment can be made for the other paths, although one would first need to evaluate the overall direction and magnitude of the effects of socio-economic variables on their descending variables.

[21] Here, it helps to consider Wright's (1921) path tracing rules for linear path models with standardized path coefficients. The association transmitted along a path is the product of the path coefficients for each link. Since all path coefficients are less than 1, even one or two weak links (say, with coefficients below 0.1) are sufficient to reduce the overall association to near zero.

adjusting for the observed set of covariates. Then, adjusting for proxies C_0^* and S^* blocks almost all the collider association so that the remaining association is reduced to a negligibly small amount. The same argument applies to the second collider path discussed here but note that the collider variable is C_0 which is activated in transmitting collider association by adjusting for its fallible measure C_0^* (but C_0's potential to induce collider association diminishes with the reliability of the observed measure C_0^*).

Note that our example of an observational study still represents a somewhat simplified scenario since treatment selection was based on only two latent variables and one observed variable. The challenge of identifying a causal effect increases when we consider a more realistic scenario, where there are other unobserved confounders U. For example, these confounders might include mental health stigma, which simultaneously makes a student less likely to attend the in-person CBT treatment (to avoid being seen in the mental health clinic and being labeled as someone with a mental illness), and which worsens depression (because of internalizing shame), $T \leftarrow U \rightarrow C_1$. In order to block these confounding effects, researchers would have to think about what these confounders are and reliably measure them. However, the brief discussion of this example demonstrates that drawing causal graphs helps researchers systematically structure and question their background knowledge about the DGP and determine which variables to reliably measure.

Randomized Experiment

In a randomized experiment, the college students are randomly assigned into the in-person CBT treatment ($T = 0$) and the online CBT treatment ($T = 1$). In other words, the decision about whether the student should receive the online vs. in-person CBT is made without any regard to their characteristics (such as C_0^*, P, S^*). Graphically, this removes any incoming arrows to the treatment indicator, meaning that C_0^*, P, S^* no longer directly affect treatment assignment. Instead, the treatment is determined only by the outcome of the random process, R, which is exogenous and unrelated to any other variable—as shown in Figure 11.

The graphical analysis indicates that without conditioning on any variables, there is 1 open causal path, 0 confounder paths and 599 closed collider paths. According to the graph, all observed baseline

variables are marginally independent of the randomly assigned treatment status, which can be empirically assessed via balance plots and tests (Imbens & Rubin, 2015). Because there are no open confounder paths, the treatment effect is identified without having to adjust for other variables. This illustrates, in graphical terms, the well-known principle that random assignment protects against confounding bias and allows the mean difference between treatment groups to provide an unbiased estimate of the average causal effect.

However, threats to validity can still occur in the context of randomized experiments, and causal graphs are able to reflect them and highlight their implications for causal inference. Because participant characteristics are not considered when determining treatment assignment, as in the initial quasi-experiment, participants may express frustration over the assignment process and change their behaviors. For example, "resentful demoralization" (Shadish et al., 2002) may occur when participants who prefer the in-person CBT are instead assigned to online CBT and, out of frustration, begin taking more antidepressants to make up for being denied their preferred treatment. Graphically, this corresponds to drawing the arrow $T \rightarrow M_1$, which adds another unintended causal effect: $T \rightarrow M_1 \rightarrow C_1 \rightarrow Y$ (as shown in Figure 12). This causal effect is not of interest to the researchers who are interested in the effect of treatment above and beyond any effects on medication use. Therefore, under this scenario, the mean difference in treatment groups is no longer an unbiased estimate of the treatment effect of interest, because this mean difference conflates two separate causal effects: the real treatment effect of interest ($T \rightarrow C_1 \rightarrow Y$) and the side effect that arises due to resentful demoralization ($T \rightarrow M_1 \rightarrow C_1 \rightarrow Y$).[22]

Suppose then that researchers attempt to identify the treatment effect of interest by excluding study participants who reported post-treatment antidepressant use from the analysis. In other words, the researchers condition their analysis on the observed variable M_1^*. Then, a graphical analysis using the

[22] Alternatively, it is also possible to represent resentful demoralization as an effect of R on M_1. In that case, it would constitute a confounding issue rather than an instance of unintended mediation. However, this makes no difference for the identification of the causal effect, because in an RCT, treatment assignment is identical to the treatment received ($R = T$).

causal DAG in Figure 12 immediately suggests that the causal effect for this subpopulation is not identified. First, because controlling for the observed proxy M_1^* is not the same as controlling for the true underlying latent variable M_1, which means that the undesired indirect effect cannot fully be controlled for. However, controlling for M_1^* also opens 1,287 collider paths because M_1^* is a descendant of the collider M_1. The open collider paths, such as $T \rightarrow M_1 \leftarrow C_0 \rightarrow C_1 \rightarrow Y$ and $T \rightarrow M_1 \leftarrow C_0 \rightarrow P \leftarrow S \rightarrow Y$, introduce spurious collider associations between the treatment and outcome that bias the estimated causal effect. The spurious collider associations persist and could even become stronger if the observed proxy were measured without error (i.e., $M_1^* = M_1$).

Now, consider an alternative scenario where although participants are randomly assigned to the in-person vs. online CBT group, participants do not always comply with their treatment assignment. Specifically, assume that some participants who take antidepressants at the baseline and are assigned to the in-person CBT treatment do not comply and succeed in receiving the online version instead. Participants may do this if they believe it is unnecessary to do both in-person CBT and take antidepressants to deal with their depression. Graphically, this corresponds to drawing the arrow $M_0 \rightarrow T$ (see Figure 13). This results in 81 open confounder paths through M_0, which means that the treatment effect is no longer identified—even if one were to condition on the fallible measure M_0^* and any other observed variables.[23] Alternatively, instead of estimating the average causal effect, the intent-to-treat effect could be estimated because the effect of R on Y remains identified even in the presence of confounding that arises due to non-compliance. Another possibility would be to use random assignment as an instrumental variable for actual treatment receipt (a graphical analysis would reveal that R is a valid instrument for T; Brito, 2010; Steiner et al., 2017).

Drawing Other Validity Threats

[23] Conditioning on the fallible measure M_0^* partially blocks the confounding association via M_0, but would also open 46 collider paths, which suggests that the treatment effect cannot be identified. However, the confounding association would be completely blocked if the observed proxy were measured without error such that $M_0^* = M_0$.

The previous paragraphs illustrated the flexibility of adapting causal graphs to different experimental contexts and validity threats that arise (Shadish et al., 2002). Here we briefly discuss a few more common validity threats. For example, in a CBT study for depression, a *history threat* might occur if an external event—such the COVID pandemic—affects participants' depressive symptoms regardless of whether they receive CBT. In the DAG, this appears as an additional cause of the post-treatment variables (history → M_1, history → C_1). This makes history an effect moderator: it changes the intervention's impact on the outcome. Any results derived from this analysis might fail to generalize to outside of the COVID pandemic.

Nonresponse and *attrition*, which result in a missing data problem, are additional examples of validity threats. Graphical models provide a powerful way to represent differential missingness processes through *m-graphs* (Mohan et al., 2013). For simplicity of exposition, assume that differential attrition results only in missing data on the depression post-test (Y), while all baseline variables are fully observed. An m-graph then shows that the actually observed outcome with missing entries, Y^m, is causally determined by the complete but latent variable Y and a separate response indicator R^Y, $Y \rightarrow Y^m \leftarrow R^Y$ (Thoemmes and Mohan, 2015). The response indicator R^Y allows researchers to explicitly represent the causes of attrition and, following our stepwise procedure, relate them to the other variables in the DAG. A graphical analysis of the m-graph can then assess whether the causal effect can be recovered from the observed data. The specific graphical criteria for the recoverability of causal effects, along with illustrative examples, are discussed in Thoemmes & Mohan (2015). However, with missing data in multiple variables, graphical recoverability analysis becomes quite complex. Finally, note that evaluating the commonly used Missing Completely at Random (MCAR) and Missing at Random (MAR) assumptions (Little & Rubin, 2019), including the appropriate selection of covariates, requires the same qualitative background knowledge about the attrition or nonresponse process as is needed for constructing an m-graph.

Interference poses a different kind of threat, arising when the treatment exposure of one individual influences the outcome of another (Ogburn & VanderWeele, 2014). Whereas standard causal

inference assumes that one individual's treatment has no impact on another individual's outcome, Ogburn and VanderWeele (2014) show how causal graphs and their analysis can be extended to represent settings where this no-interference assumption is violated. In such cases, an individual's outcome may depend not only on their own treatment but also on the treatment exposure of their peers within the same network or cluster. In a CBT study, for instance, treated participants may share coping strategies or emotional support with other treated or untreated peers, thereby indirectly affecting their peers' depression outcomes. The graphical representation of interference effects requires explicitly drawing a DAG for two individuals, say i and j, within the same peer network (e.g., duplicating the DAG in Figure 9) and then adding the interference effects as cross-unit links, $T_i \rightarrow C_{1j}$ and $T_j \rightarrow C_{1i}$. That is, treatment of individual i influences the outcome of individual j, and vice versa. A more detailed discussion of interference graphs is beyond the scope of this introduction to drawing causal graphs as one first must distinguish between different types interference patterns and, using potential outcomes notation, clearly define the average individual and indirect interference effect (for a thorough exposition, see Ogburn & VanderWeele, 2014).

Causal Graphs for Mediation Analysis

Causal DAGs are also very useful for representing and identifying mediation processes, that is, processes through which causal effects are propagated (Avin et al., 2005). Although causal mediation is its own distinct research area, we briefly address a few specific aspects about drawing causal mediation graphs and some unique, yet often overlooked, challenges in mediation analysis. For general introductions to causal mediation see, for instance, Vanderweele (2015), Hong (2015), Imai et al. (2010), Nguyen et al. (2021), Pearl (2014), or Steiner and Weinberger (2025).[24]

Drawing a causal mediation DAG follows the same six steps as outlined above. However, in Step 1, the entire causal process of interest is drawn; that is, in addition to the treatment and outcome variables,

[24] The causal mediation literature formally defines *natural* direct and indirect effects using potential (counterfactual) outcomes notation. For more realistic but complex settings in which the natural effects cannot be identified, it introduces *randomized interventional analogues*—often referred to as *interventional* direct and indirect effects (Vanderweele, 2015; Nguyen et al., 2021). However, since our article provides an entirely nontechnical introduction to drawing causal DAGs, we do not further discuss the formal definitions of these effects.

we also add the target mediators of interest and their causal interrelations to the graph. Thus, instead of a single causal path connecting the treatment and outcome, we typically have one or multiple mediator paths and a single direct causal path. Following the original causal graph in Figure 9, a research question about mediating processes might investigate whether the online CBT's impact on depression is fully mediated via CAM (C_1), or whether the direct and other indirect effects via M_1 also play a role. The causal mediation DAG in Figure 14 reflects the mediation process of interest, that is, the two target mediators M_1 and C_1 and the path-specific effects under investigation: the direct effect transmitted along $T \rightarrow Y$, and the indirect effects through $T \rightarrow M_1 \rightarrow Y$, $T \rightarrow C_1 \rightarrow Y$ and $T \rightarrow M_1 \rightarrow C_1 \rightarrow Y$.

Steps 2 and 3 then require that one draws not only the causes of treatment selection and the outcome-generating process but also the mediator-generating processes, that is, the presumed causes of each mediator. It is important to recognize that the causes of mediators often include post-treatment variables in addition to baseline variables. More specifically, for a mediator at time-point t, all causes or confounders up to time point t are relevant. The mediation DAG (Figure 14) indicates that the two mediators, M_1 and C_1, have multiple causes, with two of C_1's causes being post-treatment variables: students' physical health (H_1) and medication use (M_1) before the post-test measurements of CAM and depression.

Once all causes of the treatment, mediator, and outcome variables have been added, one can proceed with Steps 4 to 6 as before. Again, note that post-treatment variables typically confound mediator-mediator as well as mediator-outcome relationships. Focusing solely on baseline variables measured before the onset of the treatment is insufficient. Step 5 requires us to add the noisy brain scan measure C_1^* to the graph (for this example, we assume that we measured the CAM at the posttest).

Following the stepwise drawing procedure, the final mediation graph needs to correctly capture the confounding structures between the treatment, mediators, and outcome. For our two target mediators (M_1 and C_1), the final causal DAG must correctly capture the T–M_1, T–C_1, T–Y, M_1–C_1, M_1–Y, and C_1–Y confounding. Assuming that our initial DAG already included all common causes of all variables in the graph, including H_1 and M_1, no additional confounders need to be added in the mediation graph. Note that

for randomized experiments with randomly assigned treatments (T), causal mediation analysis still requires the careful consideration of M_1–C_1, M_1–Y, and C_1–Y confounding because neither M_1 nor C_1 is randomized.

The identification of direct and indirect effects rests on stronger assumptions than the identification of the total, average causal effect (Pearl, 2014; Hong, 2015; Vanderweele, 2015). For the total effect, it is sufficient to block all backdoor paths between treatment and outcome. In contrast, the nonparametric identification of direct and indirect effects requires several distinct "no unmeasured confounding" assumptions in addition to the no unmeasured T–Y confounding: (i) no unmeasured T–M_1 and T–C_1 confounding, (ii) no unmeasured M_1–C_1 confounding, and (iii) no unmeasured M_1–Y and C_1–Y confounding. Moreover, none of the mediator–outcome confounders may itself be affected by the treatment. This latter assumption is clearly violated for our target mediator C_1, where M_1 confounds the C_1–Y relationship ($C_1 \leftarrow M_1 \rightarrow Y$), but is also causally affected by T ($T \rightarrow M_1$). Thus, the indirect effect via C_1 is not identified even if all other assumptions were met. This is because identifying the indirect effect via C_1 requires controlling for the confounding induced by M_1, but doing so would also block the indirect causal path $T \rightarrow M_1 \rightarrow C_1 \rightarrow Y$. Consequently, we would fail to identify the entire effect mediated via C1. However, in such situations, one can still identify a so-called "randomized interventional analogue" of the indirect and direct effects (also called "interventional" effects; Vanderweele, 2015; Nguyen et al., 2021).

Drawing mediation graphs and identifying direct and indirect effects face two challenges that are unique to mediation or time-varying processes in general: measurement error in mediators and the representation of dynamic, time-varying mediation processes. For mediation effects to be identified, the mediators must be measured without error, since any unreliability in their measurement can attenuate the estimated indirect effects. Since neither M_1 nor C_1 is measured reliably (as shown in the mediation DAG), neither the direct nor indirect effects are identified. Even if they were reliable measured, the effects would still not be identified because of unobserved or unreliable measured variables $\{C_0, M_0, S, H_1\}$ that would be needed to block all confounding associations (see Pearl, 2014, for graphical identification criteria).

Although the mediation effects are not identified, the total effect of T on Y is identified conditional on $\{C_0^*, S^*, P\}$, as discussed above.

A second challenge arises when mediators are not static but time-varying, that is, when they evolve dynamically over time. If a mediator changes values in response to treatment exposure across multiple time points, but is measured only once, key parts of the mediation process remain unobserved. This can lead researchers to underestimate or misattribute mediation effects, since a single measurement may fail to capture the cumulative or time-varying influence of the mediator on the outcome. For instance, if medication use changes over time due to the effectiveness of CBT exposure, researchers should measure medication use multiple times and add the corresponding constructs (say, M_1, M_2, M_3, M_4), their fallible measures, and their causal interrelations to the graph. Note that it is often the case that the $M_t - Y$ relationship is confounded by treatment-induced $M_{t\text{-}1}$, $M_{t\text{-}2}$, etc. mediators, posing serious challenges for the identification of mediation effects. Even if medication use is measured only once, say M_1 or M_4, a properly drawn mediation graph still helps researchers to formulate a mediation question that can be addressed with single mediator measurement and guards against misinterpretations of direct and indirect effects. Our mediation graph in Figure 14 includes only the two mediators, M_1 and C_1, implicitly assuming that the mediation process is not time-varying or that its causal structure is such that a measurement taken just before the CAM and depression posttests validly captures the mediation effects.[25] In any case, careful consideration of the timing of mediator assessments and their reliable measurement is crucial for drawing valid inferences about mediation processes (Peters et al., 2017; Selig & Preacher, 2009).

If all causal assumptions are satisfied, researchers can estimate the direct and indirect effects using methods such as regression-based methods (VanderWeele, 2009, 2015), simulation-based algorithms (Imai et al., 2010), or weighting approaches (Hong, 2010, 2015). If the assumptions are not

[25] If the research question concerns only the total effect, then an incorrectly drawn mediation structure is inconsequential for identifying the causal effect (provided that no post-treatment variables are included in the adjustment set).

satisfied, the graphs make explicit which sources of bias remain. In this way, causal DAGs extend beyond the identification of average causal effects and provide a principled framework for analyzing mediation.

On the Credibility of Graphs and the Nonidentifiability of Causal Effects

The previous sections demonstrated how researchers can draw causal DAGs from qualitative background knowledge and then use them to check whether average causal effects are identified, which covariates to use for statistical adjustment, and how to assess the extent of remaining bias. Our proposed drawing procedure is iterative to ensure that all common causes of each variable pair in the graph are included. To obtain a causal graph that accurately represents the real DGP, one would need a fully exhaustive list of the causes of treatment selection and the outcome, as well as a large (potentially infinite) number of iterations of Step 4. Here we argue that this is not necessary. What is needed is a *credible* causal DAG that captures at least all plausible confounding sources and other validity threats (Pearl, 2014; Shadish et al., 2002). We then argue that even if credible causal DAGs for observational studies almost never allow for successful identification of a causal effect given the observed data, such graphs are still invaluable for study planning and more modest causal inference.

Credible Causal Graphs

A causal DAG is supposed to accurately represent the process that produces the observed data. However, because complete and perfect background knowledge about the DGP is rarely available, causal graphs typically fail to exactly represent the true DGP (Greenland, 2010)—except for RCTs or quasi-experiments where treatment assignment is known and fully observed and no other validity threats like non-compliance or resentful demoralization are present (as was the case for the DGPs of the quasi-experiment and the RCT). Thus, with observational data, the best one can hope for is a *close approximation* to the real DGP. Only if the approximation is sufficiently close can causal inferences derived from a DAG have real-world implications.

What is needed is a causal graph that "represents a hypothetical consensus on what is plausible and important versus that which is deemed negligible or implausible" (Pearl, 2014, p. 463). Thus, the omission of common causes or causal links (arrows) is acceptable if it is defensibly inconsequential—that

is, if it does not affect the validity of the causal conclusions about the effectiveness of an intervention (i.e., its direction and the approximate magnitude of the effect).[26] Even though the true causal DAG is rarely known with certainty, we argue that valid causal inference is still possible if the DAG is *credible* with respect to the presumed DGP—meaning that it includes all plausible and consequential common causes of variable pairs (as addressed in Step 4 and iterative Step 6).[27]

But how can we construct a credible causal DAG that is "sufficiently close" to the real DGP? At a minimum, the DAG must correctly represent the main confounding structure between the treatment and outcome. Missing minor sources of confounding—those that introduce only negligibly small associations between treatment and outcome—are often tolerable. A credible causal DAG must at least include all systematic, direct causes of the treatment selection mechanism and the outcome-generating process (Steps 2 and 3). Their causal interrelations are also crucial and must be accurately represented, including direct causal relationships and any common causes with a significant impact (Step 4). Failing to link two direct causes of the treatment or outcome by an arrow may be tolerable only if the corresponding association is arguably negligible—either because the association is weak across the full sample or only present in a small subgroup. The same logic applies to inadvertent omissions of a common cause of variables that directly determine treatment selection or the outcome. If the omitted common cause has only a minimal influence, the confounding associations transmitted along the (multiple) unrepresented paths will generally be small—and conditioning on observed variables or proxies can at least partially eliminate the corresponding noncausal associations. This reasoning applies even more strongly to unrepresented collider paths (traversing omitted common causes). When these become active through conditioning on

[26] Following de Regt's (2017) discussion of Strevens' (2008) "kairetic" account of causal explanation, it is crucial to identify the "difference makers," whereas irrelevant (inconsequential) details may be omitted. Similarly, Shadish, Cook & Campbell (2002) argue that it is sufficient to consider at least all plausible validity threats while implausible ones may be disregarded.

[27] This also implies that every (common) cause of the variables in the DAG can be linked to a well-defined intervention, such that the arrows warrant a causal interpretation. While some vagueness regarding well-defined interventions is tolerable if it is deemed inconsequential, difficulties in conceiving any meaningful intervention (e.g., on sex or race) may require adaptations of the graph—for instance, as discussed above for sex/gender (see also Hernán and Robins's (2023) discussion of "noncausal" DAGs in Chapter 9).

observed variables, they often transmit only minor collider associations—provided the omitted common causes have weak effects.

Omitting higher-order common causes (in the iterative Step 6)[28] is typically less consequential for identifying and estimating a causal effect, especially if those variables appear only on long paths because they generally transmit only weak noncausal associations and allow for blocking at multiple points (as argued above). In any case, when uncertain about a causal link or common cause between two variables, the conservative and recommended approach is to include the arrow or the common cause, respectively.

Thus, given sufficient background knowledge about the DGP, our iterative drawing procedure provides clear guidance for creating a credible and defensible causal DAG. However, as mentioned in the introductory sections, the credibility of the resulting graph strongly depends on its grounding in empirical evidence—such as prior studies, qualitative interviews with participants, expert input, or other relevant information that supports the assumed causal structure, including the absence of direct causal relationships (arrows) or variables. Importantly, the graph must represent the context-specific, local knowledge about how the data were generated. In this sense, causal DAGs are local models of local DGPs rather than idealized, abstract representations of higher-level theories (Cartwright, 2022; Maxwell, 2012). Local DGPs and their graphical representations are bound to vary across populations, settings, and time. Thus, credible DAGs offer the opportunity to convey and justify the local aspects and assumptions of the DGP in an intelligible way, allowing reviewers and readers to independently assess and critique them.

Nonidentifiability of Causal Effects

Now, assume that we succeeded in drawing a credible causal DAG for an observational study. The next challenge is that the observed covariates will almost surely be insufficient to identify the causal effect of interest (see also Greenland, 2010). Exact identification typically fails because many variables

[28] Repeated iterations of Steps 4 and 5 identify what we call "higher-order common causes," that is, variables that do not directly confound the treatment–outcome relationship but do so only indirectly, through other (common) causes already included in the graph. Because higher-order common causes appear only on longer noncausal paths, they generally transmit less—often negligible—confounding association. Thus, after two or three iterations of Step 6, any remaining omitted common causes are unlikely to pose plausible validity threats to causal inferences.

involved in the DGP are either unobserved or only unreliably measured (as in our causal DAG of the observational study). Thus, conditioning on observed covariates will not remove all confounding bias because not all confounder paths can be fully blocked, and many collider paths become partially open and induce spurious collider associations. Note that drawing an unrealistic DAG that allows for effect identification with the observed data but is misaligned with the real DGP (e.g., by omitting relevant common causes or ignoring distinctions between latent variables and unreliable measures) is not a solution but self-deception. Because such a graph deviates from the real DGP, the resulting effect estimates and causal claims are highly speculative and likely misleading with regard to corresponding real-world interventions.

Given that credible causal graphs typically result in negative identification results, one may ask why researchers should even make the effort to draw a causal graph. There are several reasons why creating a credible graph is not only useful but essential for causal inference—even when exact identification is not achievable.

First, DAGs allow researchers to systematically organize and formalize context-specific qualitative background knowledge about the presumed DGP that gave rise to the observed data. The DAG explicitly shows the main causal assumptions on which causal inference rests and allows researchers to assess their credibility. It is important to recognize that the Rubin Causal Model (or a corresponding econometric model; Heckman & Pinto, 2022) also requires the same qualitative background knowledge to select covariates that ensure unconfoundedness. The difference is that the Rubin Causal Model does not provide tools or a language to formalize this knowledge or reliably select a valid adjustment set (Steiner et al., 2023). The lack of clear guidance in assessing unconfoundedness invites researchers to claim that observed baseline covariates (approximately) meet the assumption—often without justification. Causal graphs, in contrast, force researchers to make key assumptions explicit and defend the graph they draw.

Second, if a causal DAG is drawn before data collection, it provides clear guidance about which variables need to be reliably measured. If the graph is constructed after the data have been collected, it

still helps assess whether the available data will remove most of the confounding bias. Moreover, a published DAG provides a solid foundation for improved replication studies.

Third, when the causal effect is not identified, a graphical analysis of open confounder and collider paths is valuable for assessing the direction and approximate magnitude of remaining confounding bias (as noted above). Such an assessment requires additional information about the DGP that is not directly encoded in DAGs, specifically the signs and approximate strengths of causal links (i.e., arrows). However, graphs and graphical analyses can be extend to partially reflect corresponding information (VanderWeele & Robins, 2010). Such graphical analyses may also inform sensitivity analyses that assess the robustness of effect estimates to unobserved confounding. However, research and formal guidance in this area are still limited.

Fourth, if observed covariates fail to block all confounding associations, the causal DAG may suggest alternative identification strategies. For instance, the average causal effect might be identifiable via the front-door criterion or, more generally, via the *do*-calculus (Pearl, 2009). Given a DAG, the *do*-calculus can be implemented in software like R (Tikka & Karvanen, 2017), which returns a nonparametric estimator if identification is possible. Additionally, the DAG may reveal an instrumental variable—possibly conditional on other covariates—that can be used to identify the complier average treatment effect (Angrist et al., 1996; Sheehan et al., 2024; Steiner et al., 2017). Similarly, if a pretest measure of the outcome is available (as in our example), difference-in-differences methods may be applicable if the common trend assumption holds (Lechner, 2011; Kim & Steiner, 2021b). In both cases, whether key assumptions are approximately met and which covariates to control for can more reliably be assessed with the help of causal DAGs (Sheehan et al., 2024).

Fifth, credible causal DAGs can indicate when observed data are clearly insufficient for causal inference, suggesting researchers should refrain from causal interpretations of effect estimates. This is especially important when major confounders are unobserved or poorly measured, or when data are affected by endogenous sample selection bias (due to conditioning on post-treatment colliders; Elwert &

Winship, 2014; Cunningham, 2021). In such cases, associational or disparity-based interpretations are still valid (Greenland, 2010).

Conclusion

We introduced a stepwise procedure to support researchers in drawing credible causal DAGs that represent highly contextualized DGPs. A summary of important considerations when drawing causal DAGs is presented in Box 1. The graphs can represent not only confounding issues but also many types of validity threats to RCTs and quasi-experimental designs, including attrition and nonresponse issues, interference effects, noncompliance, differential maturation, history effects, compensatory equalization, resentful demoralization, or endogenous sample selection issues (Elwert & Winship, 2014; Matthay & Glymour, 2020; Ogburn & VanderWeele, 2014; Reichardt, 2019; Shadish et al., 2002; Sheehan et al., 2024; Thoemmes & Mohan, 2015). Though we briefly discussed how to draw some of the validity threats, further research is necessary in translating all possible threats to graphical representations in causal DAGs.

Although causal DAGs are flexible and can be adapted to different research contexts, it is important to remember that DAGs are formal mathematical objects with an underlying structural causal model that links the graph to probability theory (Pearl, 2009). As such, causal DAGs are subject to specific constraints. For example, the DAG must be acyclic, meaning that it must be impossible to start at any node and follow a sequence of directed edges that loops back to the starting node. Similarly, analytic variables like respondent ID or time of repeated measurements or longitudinal data are typically not included in a DAG—instead, the corresponding respondent characteristics and time-indexed variables are represented. In short, researchers are not free to draw just anything; the DAG must mirror the real DGP in a meaningful, coherent way. In a similar vein, DAGs do not capture all the assumptions required for valid causal inference. For instance, the positivity assumption, that treatment and control groups have overlapping support in the joint covariate distribution, or the invariance assumption, that individuals do not change their behavior in response to a conceivable, real-world intervention, are not encoded in DAGs, which only represent how the observed data were actually generated.

Causal inference without context-specific qualitative background knowledge about the DGP is impossible, regardless of whether one relies on the Rubin Causal Model, Campbell's validity typology and threats, or causal DAGs. However, the advantage of causal graphs is that they provide an intuitively understandable yet rigorous "language" to formalize, communicate, and explicitly discuss the necessary causal background knowledge. Most importantly, because DAGs are intelligible, causal representations of the real DGP, they can be *used* to formally define causal effects, derive testable implications, determine whether a causal effect is identified and which covariate sets remove the confounding bias, and help evaluate the direction and potential magnitude of any remaining bias.

Overall, causal graphs may help establish a broader culture of making the qualitative assumptions underlying causal claims explicit and open to critical evaluation—rather than pretending that the application of "causal methods," such as propensity score matching or inverse-probability-of-treatment weighting, to a "sufficiently rich" dataset successfully eliminates confounding bias without ever justifying that the observed covariates satisfy the required assumptions (Greenland, 2010). The same caution applies to causal inference methods that rely on artificial intelligence and machine learning, such as double/debiased machine learning (Chernozhukov et al., 2025). These methods do not guarantee valid causal inference unless the set of observed covariates satisfies the adjustment criterion—which can only be assessed based on qualitative background knowledge. Without this assurance, no statistical or computational approach—however advanced—can compensate for the absence of a credible causal model. Therefore, even though DAGs may never fully capture the true DGP and observed data usually fall short of identifying a causal effect, using DAGs is an essential first step toward more structured, transparent, and honest reasoning about causal assumptions.

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Table 1.
Steps in drawing a causal graph

Step	Description	Questions to Consider
1. Causal Process	Add the treatment and the outcome, and draw a direct effect from the treatment to the outcome. Add variables that mediate the cause/treatment and the outcome (skip if mediators are not of interest).	What is the treatment and control contrast? What are the causal processes (i.e. mediators) that leads to the treatment and control groups to differ in terms of the outcome? If mediator(s) are included, is the relationship between the treatment and the outcome fully explained by the mediator(s)?
2. Selection Process	Add variables that determine the assignment mechanism (in an experiment) or the selection mechanism (in a non-experiment), and draw a corresponding direct effect into the cause/treatment.	What variables make a participant more likely to select into the treatment (vs. control) group or vice versa? If selection is based on latent constructs (e.g., self-selection based on ability) instead of observed ones (e.g., administrator selection on reported achievement scores), are those constructs reliably measured?
3. Outcome-generating Process	Add variables that determine the outcome, and draw a corresponding direct effect into the outcome.	What (baseline) variables influence the value of the outcome? Do they affect the outcome directly or indirectly through the outcome's mediator(s)?
4. Pairwise Interrelations	Check each pair of variables that are not directly connected, and draw direct effects for pairs that should be directly causally related. Add *all* common causes that produce spurious associations between the two variables.	What is the direction of the causal relationship? If the variables are already connected by a chain of mediators, should there be a direct effect that operates above and beyond the effect of the mediator? Is there a common cause between two variables, and if so, what are the common causes?
5. Additional Observed Variables	Add any relevant observed variables in your dataset that have not yet been included in the causal graph.	Do the observed variables mediate any of the pairs of variables that are causally connected? Are the observed variables idiosyncratic causes of single variables? Are the observed variables common causes of pairs of any pair of variables? Are the observed variables fallible measures of the underlying construct?
6. Iterate steps 4 & 5	For any variables that were added in steps 4 or 5, repeat step 4 (check pairwise relations) until no additional variables are drawn.	

Box 1. **Key considerations for drawing causal graphs.**

Causal relation(s) of interest. Clearly depict the treatment (cause) and outcome of interest, along with their presumed direct and indirect causal pathways. If mediation is relevant, include mediators that plausibly transmit the causal effect.

Effect moderators. Since graphs are nonparametric representations of the data-generating process, moderator and interaction effects are automatically encoded—thus, they are not explicitly drawn.

Observed and unobserved variables. Include all variables—both observed and unobserved—that are part of the local, context-specific data-generating process.

Measurement error. Differentiate between unobserved latent constructs and their imperfect (unreliable or biased) observed indicators (Greenland, 2010; Steiner & Kim, 2016).

Common causes. Include all plausible common causes (confounders) of any pair of variables. If in doubt, always include a common cause, even if unknown.

Arrows (causal links). Connect any pair of variables with an arrow to represent their direct causal relation (i.e., a real-world intervention on the presumed cause would affect or has at least the potential to affect the other variable). When uncertain, err on the side of including arrows rather than omitting them.

Temporal ordering. Respect the temporal sequence of measured variables. A variable measured earlier in time can only be a cause—not an effect—of variables measured later. Where possible, reflect temporal flow visually, such as by orienting the graph from left (past) to right (future).

Time. Time-varying variables and effects over time should be drawn using time-indexed variables. Time itself is not drawn as a variable in the graph.

Clustered/hierarchical data structures. Graphs representing data-generating processes that involve cluster-level variables should explicitly include those variables rather than a generic cluster ID indicator.

Data restrictions. Include variables that influence sample inclusion or exclusion, particularly if these are post-treatment variables (Elwert & Winship, 2014; Cunningham, 2021).

Aspects of the research design. Explicitly represent design elements—such as blocking, matching, stratification, counterbalancing, randomizations, or eligibility criteria—as distinct variables when they influence the data-generating process or analysis (Mansournia et al., 2013; Shahar & Shahar 2012; Steiner et al., 2017).

Treatment fidelity and contamination. Explicitly draw measurement error in the treatment variable or treatment fidelity (implementation) issues. Represent treatment contamination and interference/peer effects (Ogburn & VanderWeele, 2014; Vanderweele, 2022).

Validity threats. In addition to confounding due to differential treatment selection, represent other plausible threats to internal validity. These may include maturation, history, testing effects, instrumentation changes, attrition, nonresponse, compensatory equalization, or resentful demoralization

Figure 1

Step 1. Causal Process

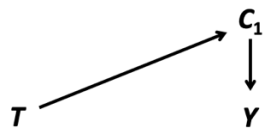


Figure 2

Step 2: Selection Process. Newly added variables and arrows are set in bold.

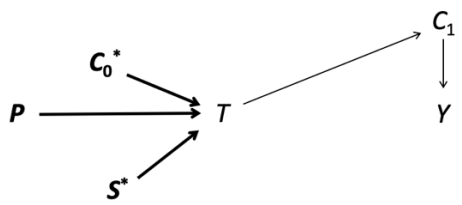


Figure 3

Step 3: Outcome-Generating Process. Newly added variables and arrows are set in bold.

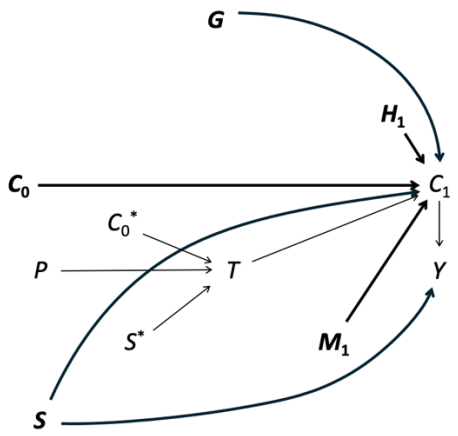


Figure 4

Step 4(a): Direct Causal Relations. Newly added variables and arrows are set in bold.

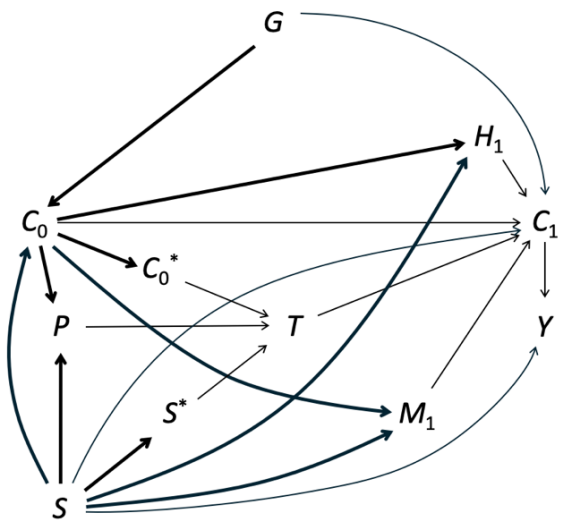


Figure 5

Step 4(b): Confounding Relations. Newly added variables and arrows are set in bold.

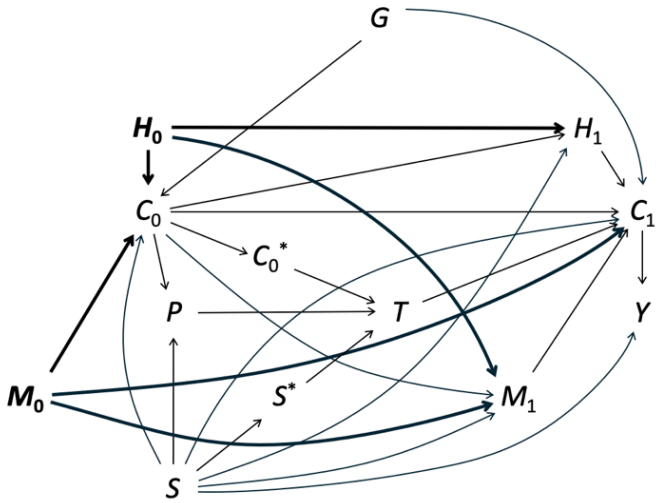


Figure 6

Step 5: Additional Observed Variables. Newly added variables and arrows are set in bold.

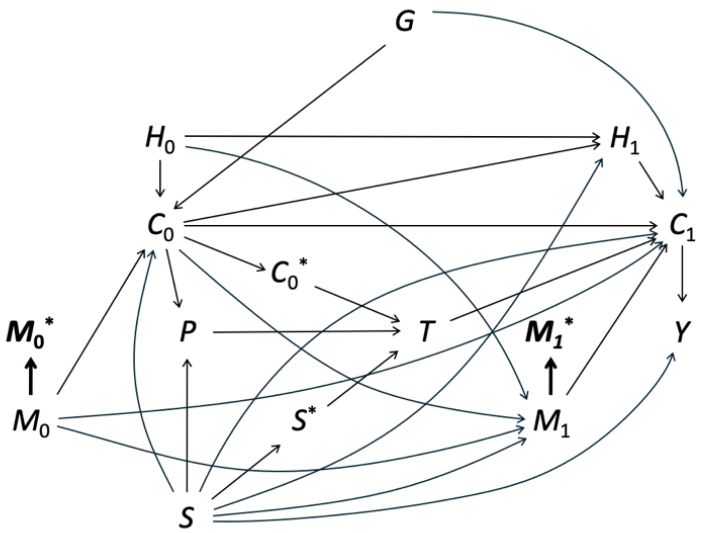


Figure 7

Step 6: Additional Direct Causal Relations. Newly added variables and arrows are set in bold.

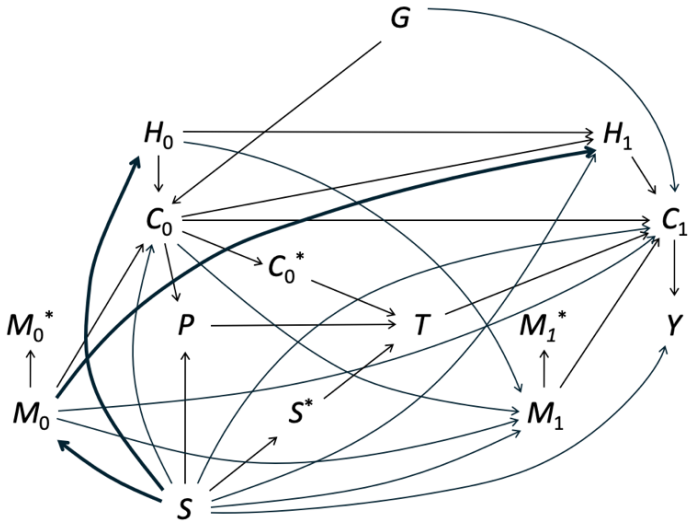


Figure 8

Step 6: Additional Confounding Relations. Newly added variables and arrows are set in bold.

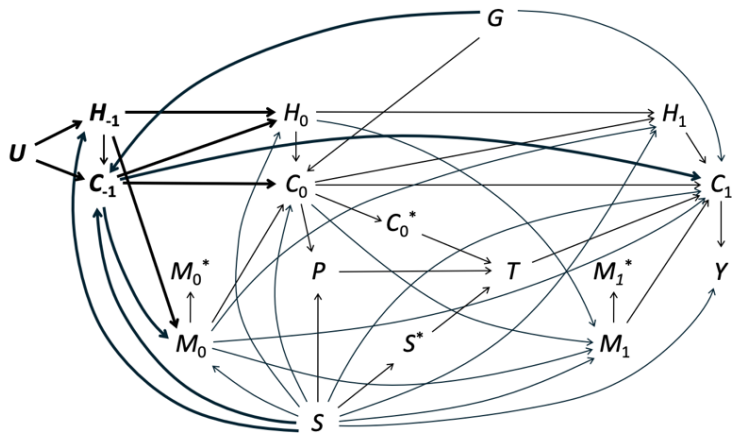


Figure 9

Final Graph of the Quasi-Experimental Study

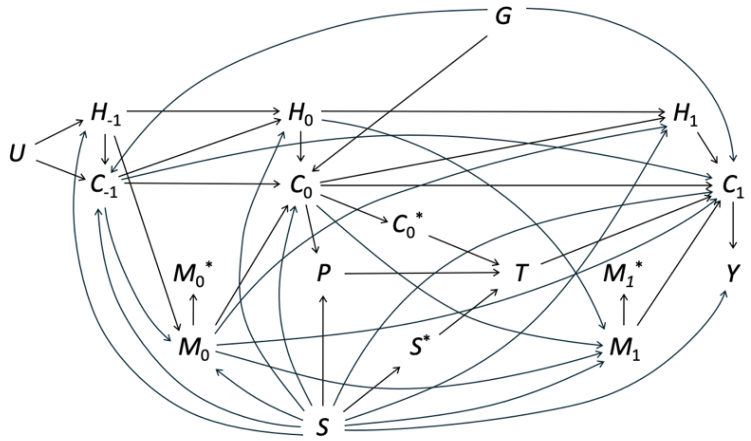


Figure 10

Causal Graph of the Observational Study

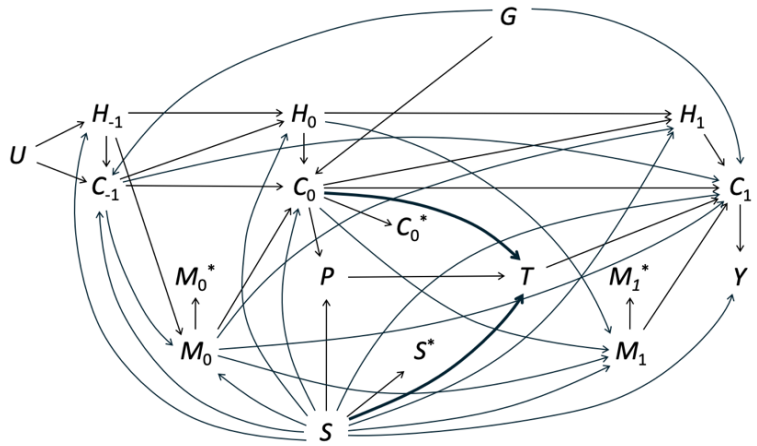


Figure 11

Causal Graph of the Randomized Experiment

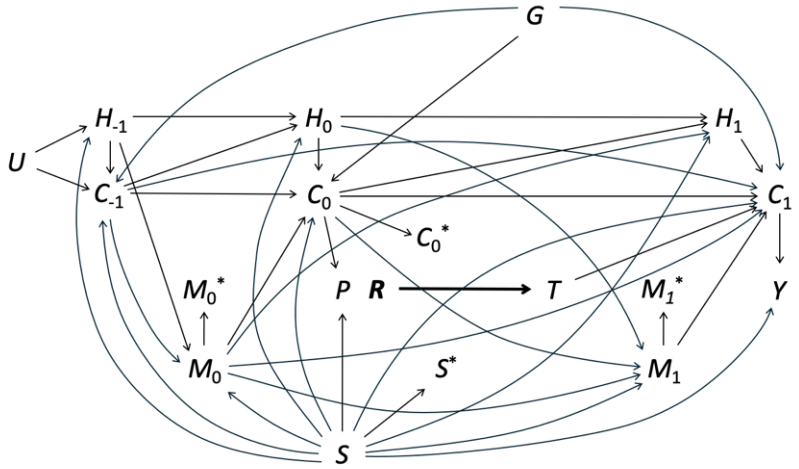


Figure 12

Causal Graph of the Randomized Experiment with Resentful Demoralization

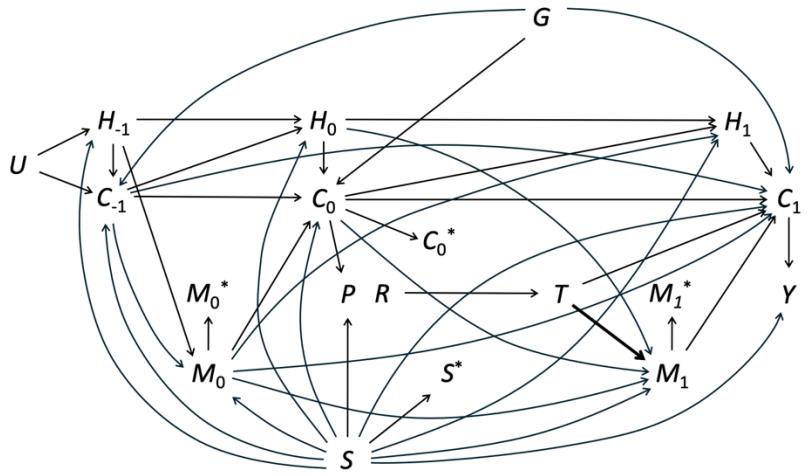


Figure 13

Causal Graph of the Randomized Experiment with Treatment Non-Compliance

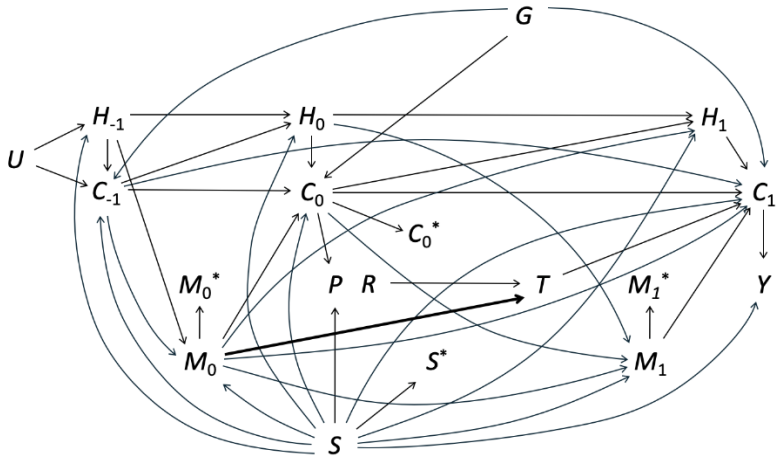
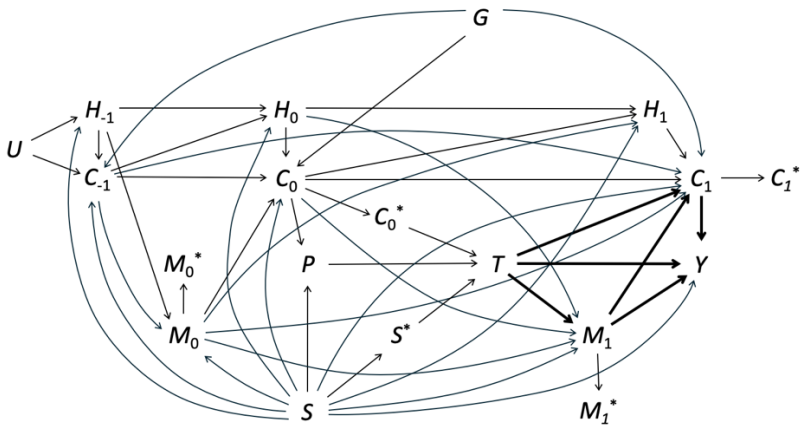


Figure 14

Causal Graph of the Mediation Graph. The direct effect and indirect effects through C_1 and M_1 are set in bold.



Appendix

A Brief Introduction to Causal Directed Acyclic Graphs

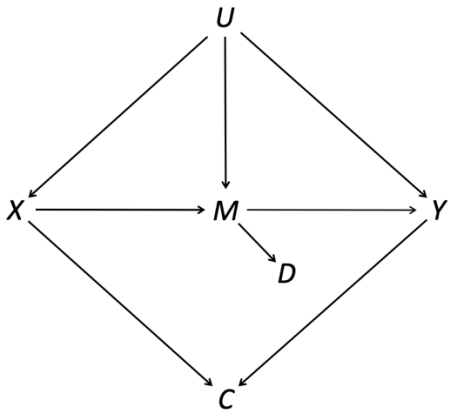
What follows is a brief introduction to causal directed acyclic graphs (causal DAGs), which are commonly used in causal inference. We restrict this introduction to graphical terminology and graphical models, without linking them to formal probability theory (which would require introducing either nonparametric, structural causal models or the causal Markov assumption). Thus, probabilistic and statistical concepts, including (nonparametric) identification formulas or estimators, are not discussed here. For more comprehensive introductions, see, for instance, Elwert (2013), Feng (2024), Hernán & Robins (2023), Huntington-Klein (2022), Morgan and Winship (2015), Pearl (2010a), Pearl et al., (2016), or Steiner et al. (2023).

Basic Definitions

Causal graphs are formal objects that are used to encode a researcher's qualitative background knowledge and assumptions about a data-generating process (DGP) under investigation. More formally, they represent the structure of causal relationships among a system of variables. A toy-example of a causal graph with six variables is shown in Figure A1. *X* might represent a treatment of interest, *Y* the outcome measure, and *M* a mediator variable. *U* is confounder of the causal *X*-*Y* relationship, *C* is a (collider) variable that is affected by the treatment and outcome, and *D* is a variable descending from *M*.

Figure A1.

Generic Causal Graph



Variables in the graph are connected by directed arrows, which indicate a causal relationship in the direction of the arrow. The arrow in the causal link $X \rightarrow M$, for instance, signifies a causal relation in the sense that an actual, real-world intervention on X would have (or at least allows for) an impact on M. On the other hand, intervening on M would have no impact on X. Any sequence of arrows between two variables that does not go through the same variable more than once is called a *path*. For example, there are exactly five paths connecting the two variables X and Y: (1) $X \rightarrow M \rightarrow Y$, (2) $X \rightarrow C \leftarrow Y$, (3) $X \leftarrow U \rightarrow Y$, (4) $X \rightarrow M \leftarrow U \rightarrow Y$, and (5) $X \leftarrow U \rightarrow M \rightarrow Y$.

These paths can be distinguished into three types depending on the direction of the arrows along the path:

- A *causal path* is a path where all arrows point in the same direction. Any intermediate variables along the path are called *mediating variables*. Path (1) $X \rightarrow M \rightarrow Y$ is a causal path, where M is a mediating variable.
- A *collider path* is a path that contains at least one variable with two arrows pointing towards it (i.e., colliding in it). Such variables are called *collider variables*. Paths (2) $X \rightarrow C \leftarrow Y$ and (4) $X \rightarrow M \leftarrow U \rightarrow Y$ are collider paths, where C is a collider variable in path (2) and M is a collider variable in path (4).
- A *confounder* or *confounding path* is any path that is not a *causal* or *collider* path. Confounding paths contain no collider variables and exactly one variable with two outgoing arrows. Such a variable is called a *confounder* (or common cause). Paths (3) $X \leftarrow U \rightarrow Y$ and (5) $X \leftarrow U \rightarrow M \rightarrow Y$ are confounder paths, where U is the confounder in both path (3) and (5).

Paths are defined for any pair of variables and can have any length so long as they do not go through the same variable twice. The same variable may play a different role depending on the path being considered. For example, although M is a mediator along path (1) $X \rightarrow M \rightarrow Y$, it is a collider along path (4) $X \rightarrow M \leftarrow U \rightarrow Y$. Note that although U is a confounding variable in path (4) $X \rightarrow M \leftarrow U \rightarrow Y$, path (4) is still a collider path and not a confounder path because the path has a collider variable, M.

A graph in which all the arrows are directed (no bi-headed edges) and no causal path forms a cycle that leads back to the starting variable is called a *directed acyclic graph* (DAG). If the variables drawn in the DAG correspond to a real system of variables, and (i) the arrows imply causal relationships as defined above and (ii) no common causes (confounders) between any pair of variables have been omitted from the graph, then the graph is called a *causal* DAG. Note that for an arrow to imply a causal relationship, one must be able to conceive of an intervention (at least hypothetically) on the cause—a condition that can be difficult to meet for some variables. For example, it is not immediately clear what an intervention on sex or race would entail, or whether the measured variable would truly represent the underlying construct (for a more formal definition and discussion of causal DAGs, see Hernán & Robins, 2023). Accordingly, DAGs that do not satisfy these two conditions are not causal DAGs and cannot be used to draw valid inferences about real-world interventions, unless the violations of these conditions are defensibly inconsequential (see the discussion on "credible" causal DAGs in the article). Figure A1 is only a toy example, and causal DAGs in practice are rarely as simplistic (see the CBT graph in Figure 9 of the article for an attempt at a more realistic graph). However, for the sake of explaining the key concepts behind causal graphs, we will presume that Figure A1 represents a causal DAG.

Associations Transmitted along Paths

In causal DAG, all statistical associations between pairs of variables are transmitted only through the paths that connect them. If the paths do not correctly capture all associations, the DAG is not a causal DAG. The flow of association transmitted in a causal graph can be thought of as a chain of whispers, similar to the children's game "Telephone." For the variables X and M, we can think of the link $X \rightarrow M$ as X "whispering" a message to M. Thus, X and M are associated because M receives X's message. Depending on the strength of the causal association, the message may be only partially received or understood—due to the background noise, that is, the idiosyncratic causes or errors that are, by convention, not explicitly shown in a graph.

Not all paths transmit association. Whether a path containing more than two variables does or does not transmit association depends on the direction of arrows (whispers) that connect the variables on the path—or in other words, it depends on the type of path:

- Causal paths transmit causal association: Causal paths like (1) $X \rightarrow M \rightarrow Y$ transmit an association because X whispers to M, and M whispers to Y. The association is causal, because if we were to intervene on X (i.e., force X to send a specific message), the message would be propagated to Y (through M).
- Confounder paths transmit noncausal (confounding) association: Confounder paths like (3) $X \leftarrow U \rightarrow Y$ also transmit an association because the confounder U simultaneously whispers the same message to X and Y, making them associated. The association is noncausal because intervening on X would not transmit an association to Y along this path, since X is not causally connected to Y.
- Collider paths do *not* transmit any association: Collider paths like (2) $X \rightarrow C \leftarrow Y$ do not transmit association because both X and Y whisper independently to C, but C remains muted on this path. Thus, C absorbs the whispers, and no association is transmitted between X and Y.

Thus, the association between X and Y in our causal graph (Figure A1) is transmitted along three of the five paths: the causal path (1) $X \rightarrow M \rightarrow Y$ and the confounder paths (3) $X \leftarrow U \rightarrow Y$ and (5) $X \leftarrow U \rightarrow M \rightarrow Y$. The goal of causal inference is to isolate the causal association from the confounding associations. This can be achieved by preventing, through appropriate statistical adjustment, the confounder paths from transmitting noncausal associations.

Blocking Paths via Conditioning

A path is *open* if they transmit causal or noncausal association (whispers), and *blocked* or *closed* if no association is transmitted. By default, causal and confounding paths are open whereas collider paths are blocked. The causal effect of X on Y can be identified if we succeed in blocking all noncausal paths while keeping the causal paths open. Noncausal paths can be blocked by conditioning on variables, that is, by covariate adjustment methods such as regression analysis, propensity score matching, or inverse-probability-of-treatment weighting methods (Imbens & Rubin, 2015; Schafer & Kang, 2008; Steiner &

Cook, 2013). However, conditioning on variables may inadvertently block causal paths and open collider paths that were previously naturally blocked. Thus, we need to carefully distinguish between the different path types when assessing the impact of covariate adjustments.

- Causal paths: Conditioning on at least one variable along a causal path will close the path and prevent the transmission of any causal association along the path. Conditioning on a *descendant* (a variable influenced by a variable on the path) partially blocks the causal association from being fully transmitted because descendants carry information similar to what Y receives, which means controlling for them removes part of the association transmitted to Y.
- Confounder paths: Conditioning on at least one variable along a confounder path will close the path by preventing any association along the path from being transmitted. Conditioning on a *descendant* of a variable on the path partially blocks the confounding association since descendants carry information similar to what Y receives, which means controlling for them removes part of the association transmitted to Y.
- Collider paths: Conditioning on all the collider variables along a collider path, and none of the non-collider variables, will open the collider path. For instance, consider the collider path $X \rightarrow C \leftarrow Y$. Then, conditioning on C unmutes the collider, that is, learning what it had heard and what X had whispered allows us to infer Y's whispers. Thus, X and Y become associated along the collider path. Conditioning on a descendant of a collider has the same effect since the descendant carries information similar to the collider. The association induced by an open collider path is called *collider bias*.

Now we can summarize the concept of blocking paths in a single, formal statement (Pearl, 2009; Pearl et al., 2016). A single path from X to Y is blocked by a set of variables **V** if the path contains a variable S that is (i) a mediating variable ($\rightarrow S \rightarrow$) or a confounder ($\leftarrow S \rightarrow$) *and* S is included in **V**, *or* (ii) is a collider ($\rightarrow S \leftarrow$) *and* neither S nor a descendant of S is included in **V**. Thus, multiple sets may successfully block a single path.

***d*-Separation**

The concept of *d*-separation extends the idea of blocking a single path from *X* to *Y* to blocking *all* paths connecting *X* and *Y* (Pearl, 2009; Pearl et al., 2016). If a variable set **V** simultaneously blocks *every* path between variables *X* and *Y*, then *X* and *Y* are *d*-separated conditional on **V**. For instance, in our causal DAG (Figure A1), *X* and *Y* are *d*-separated conditional on **V** = {*U*, *M*} because **V** blocks all five paths connecting *X* and *Y*. {*U*, *M*} is the smallest possible conditioning set but **V** = {*U*, *M*, *D*) also blocks every path (note that *C* cannot be included in **V** because this would induce collider bias). *d*-separation is typically used to derive testable implications from the causal graph, that is, if two variables are *d*-separated in the DAG then they must be probabilistically independent conditional on **V** (provided the DAG correctly reflects the DGP).

Adjustment Criterion

To assess whether a set of observed covariates can identify the causal effect of *X* on *Y* given the causal graph, we can apply the adjustment criterion (Shpitser et al., 2010; or the slightly more restrictive backdoor criterion, Pearl, 2009). The average causal effect between two variables *X* and *Y* is nonparametrically identified if an adjustment set of observed variables **V** exists, such that

- **V** blocks all noncausal (confounding and collider) paths from *X* to *Y*, and
- no variable in **V** is on the causal path from *X* to *Y*, or descends from a causal path from *X* to *Y*.

Given a positive identification result, the adjustment set is then used for statistical adjustments that control for **V** in a regression model, matching on **V**, or use **V** to estimate the propensity score for inverse-probability-of-treatment weighting (Imbens & Rubin, 2015; Schafer & Kang, 2008).

Applying the adjustment criterion to our causal DAG in Figure A1 reveals that the causal effect of *X* on *Y* is identified and that only a single adjustment sets, **V** = {*U*}, blocks all noncausal associations—provided *U* is observed and reliably measured. Including *M* in the adjustment set would block the causal path, $X \rightarrow M \rightarrow Y$, and thus remove the entire causal effect of interest. Including *D* would also partially block the causal path, because *D* is a descendant of the mediator *M*. The post-outcome variable *C* cannot be included either because it would open the collider path, $X \rightarrow C \leftarrow Y$, and introduce spurious collider bias. Given a causal DAG, the adjustment criterion can be applied using

statistical software packages like *dagitty* in R (Textor et al., 2016). The challenge is therefore not how to identify an appropriate covariate set once a causal DAG is drawn, but rather how to draw such a graph in the first place.